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Using Construction-Integration Theory to Interpret Reading Comprehension Instruction for Students with Autism Spectrum Disorder: A Systematic Review and Meta-Analysis

Shuai Zhang
Department of Education, School of Arts, Sciences, Business, and Education, University of Saint Joseph (Connecticut), West Hartford, Connecticut, USA

Debra A. Prykanowski

David A. Koppenhaver
Reading Education and Special Education Department, Reich College of Education, Appalachian State University, Boone, North Carolina, USA

ABSTRACT

More than 800,000 students with autism spectrum disorder (ASD) are served in U.S. public schools, a number that has increased annually for almost 20 years since Individuals with Disabilities Education Act (IDEA) included ASD as an eligibility category. These students often experience widespread and persistent reading comprehension challenges. A systematic review and meta-analysis, anchored in a construction-integration model of reading comprehension, was undertaken to examine single-subject research design (SSRD) and group design studies aimed at remediating reading comprehension challenges in students with ASD. Reading comprehension instruction was operationalized by surface, textbase, and situation model levels. Random effect modeling with robust standard error estimations suggested small yet statistically significant intervention effects. Situation model-level interventions demonstrated the highest intervention effects in group design studies. Across designs, retelling and curriculum-based question-answering measures demonstrated larger effect sizes than cloze and norm-referenced measures. Methodological quality analysis suggested that few individual SSRD and group studies had high design quality with large intervention effects. We concluded that reading comprehension instruction focused on the situation model level should be prioritized for students with ASD, and future studies should reduce bias in designing intervention studies.

Autism spectrum disorder (ASD) is characterized by significant challenges with communication, behavior, and social skills, as well as frequent learning challenges (American Psychological Association, 2017). One in 160 school-aged children worldwide is diagnosed with ASD (World Health Organization, 2019), and U.S. public schools currently serve more than 800,000 students with ASD (US Department of Education, 2022). With the passage of the *Every Student Succeeds Act* (, 2015), special education and general education teachers are required to operate from the premise that all students can succeed academically and to use evidence-based practices to achieve this end (Chenoweth, 2016). An ongoing challenge in meeting this legal mandate for students with ASD has been the learning differences and persistent reading comprehension challenges of students with ASD (American Psychological Association, 2017; CDC, 2020) within the context of one-size-fits-all instructional approaches lacking differentiation (Bondie et al., 2019).

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Characteristics of Students with ASD

ASD is a neurodevelopmental disorder characterized as a spectrum because of the wide-ranging nature and varying severity of symptoms (American Psychological Association, 2017). The noted challenges with communication, behavior, and social skills pose numerous obstacles for educators. Communication challenges may include, but are not limited to, impaired speech, an inability or unwillingness to use speech for communication, or the repetition of words and phrases spoken by others (CDC, 2020; National Research Council, 2001). Students with ASD may find it difficult to express their thoughts, engage in turn-taking, interpret facial expressions, or consider others' perspectives (American Psychological Association, 2017). These factors can contribute to miscommunications in classroom interactions with peers and teachers (Sigafoos et al., 2006). Behavior and social skill challenges are widespread yet vary in degree and nature (CDC, 2020). Students with ASD may respond unpredictably to sensory input, engage in repetitive behaviors, seek unchanging routines, or struggle with activity transitions (American Psychological Association, 2017; CDC, 2020). Challenging behaviors, especially those that escalate in severity and frequency, can interfere with academic learning, disrupt classroom activity, decrease the quality of human interactions, and even present risk of injury (Sigafoos et al., 2006). In social situations, students with ASD may avoid eye contact or fail to perceive social cues, complicating the possibility of successful participation in reading lesson interactions or informal conversations. These communication, behavior, and social skill differences may pose challenges when teaching students with ASD to read (Ricketts et al., 2013).

Reading Comprehension in Students with ASD

Although there are substantial individual differences, students with ASD tend to exhibit relative strengths in word identification and reading fluency but relative weakness in reading comprehension (see Brown et al., 2013, for meta-analysis). Some scholars attribute the comprehension challenges of students with ASD to limited world experiences and background knowledge (Wahlberg & Magliano, 2004); however, most researchers attribute these challenges to broader underlying oral language impairments (Nation et al., 2006; Ricketts et al., 2013). Students with ASD struggle in general with meaning, or semantic knowledge (Brown et al., 2013), particularly in comprehending figurative language (Norbury, 2005). A few studies have focused on inferencing as a challenging aspect of reading comprehension for students with ASD (Randi et al., 2010). Saldaña and Frith (2007) reported that adolescents with ASD could make bridging inferences in short texts but hypothesized that the problem might be more specific to creating mental representations, or situation models (Kintsch, 1988; Kintsch & Rawson, 2005), for more complex texts. Students with ASD may find it particularly challenging to develop mental representations for narrative text because they struggle to understand the perspectives of different characters (Garcia-Perez et al., 2008).

Researchers in ASD have proposed three broad theoretical explanations for reading comprehension difficulties in students with ASD: Theory of Mind (ToM) deficits, executive function (EF) deficits, and central coherence (CC) deficits. ToM is the understanding that each individual has unique thoughts and has been widely identified as a source of struggles in making inferences and predictions and in understanding character motivation and actions in narratives (Carnahan & Williamson, 2010). EF is an umbrella term referring to a wide range of cognitive functions (e.g., working memory, updating, planning, shifting, inhibition), potentially explaining challenges in reading purposefully, self-monitoring comprehension during reading, and developing coherent understandings of text (Carnahan & Williamson, 2010; Ozonoff et al., 2007). CC deficits are tendencies to focus attention on details rather than important ideas or overall meaning (Frith, 1989). CC deficits are proposed as the reasons some individuals with ASD find it difficult to summarize key details, identify a gist, or relate supporting details to main ideas (Carnahan & Williamson, 2010; Martin & McDonald, 2004). These cognitive deficit theories have formed the basis of attempting to understand reading comprehension difficulties in ASD. However, few meta-analyses have examined types of reading instruction that may influence reading comprehension challenges in students with ASD. The current research study aims to inform instructional and research design by reviewing reading comprehension instruction for students with ASD from a construction-integration (C-I) model perspective (Kintsch, 1988, 1998, 2019). The C-I model proposes that successful reading comprehension requires the reader to construct a mental representation of a text at three levels, surface (i.e., the actual words of the text), textbase (i.e., a set text-based propositions and their relationships), and situation models (i.e., information from the text integrated with the reader's world knowledge).

Previous Reading Comprehension Instruction Reviews

Previously, researchers conducted several reviews of effective reading comprehension instruction methods for students with ASD. Three reviews focused on broader reading skills but also included reading comprehension. Chiang and Lin (2007) reviewed sight word knowledge

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and reading comprehension instruction for students with ASD published between 1986 and 2006. They identified four studies focused on reading comprehension and concluded that successful interventions included cooperative learning, peer tutoring, anaphoric cueing, and prereading questions. Whalon et al. (2009) reviewed how phonemic awareness, phonics, fluency, vocabulary, and reading comprehension were fostered during instruction for students with ASD. They identified five studies in which reading comprehension interventions were effective, with particular success demonstrated when peer-mediated strategies (e.g., reciprocal teaching, class-wide peer tutoring) were included. More recently, Bailey and Arciuli (2020) reviewed 19 studies that addressed the National Reading Panel (NRP)-supported reading components, including five studies published between 2011 and 2017 that included reading comprehension instruction. They found that instruction that addressed multiple NRP components and was conducted with high methodological quality was associated with gains in fluency and reading comprehension. Although these researchers reviewed reading instruction components delivered to students with ASD, reading comprehension instruction was not the only focus in these studies. As a result, it was difficult to understand the level of comprehension addressed in intervention studies and whether higher-level comprehension intervention (e.g., situation level) was more effective than lower-level comprehension intervention for students with ASD.

Six reviews focused only on reading comprehension instruction. Senokossoff (2016) synthesized 12 studies from 1990 to 2012 examining reading comprehension strategies for students with high-functioning ASD. Senokossoff found that specific strategies (e.g., using prompting questions to guide the thinking process) and general instructional methods (e.g., peer-tutoring, small-group guided reading) both showed potential to improve reading comprehension. Finnegan and Mazin (2016) reviewed 15 experimental intervention studies from 1989 to 2015 for students with ASD. They reported that direct, guided reading comprehension instruction and graphic organizer strategies showed more promising results than cooperative learning, anaphoric cueing, and question generation. However, neither of the studies included methodological quality indicators, making it difficult to determine a functional relation between these practices and the reading comprehension outcomes.

El Zein et al. (2014) synthesized 12 experimental, quasi-experimental, single subject, and pre–post-design studies from 1980 to 2012 reporting reading comprehension outcomes for K–12 students with ASD. They analyzed the instructional strategies used and their effects on reading comprehension from a cognitive deficit theories' perspective (e.g., ToM, EF, CC). Seven of the 12 studies were not identified in previous reviews and suggested additional effective practices including using explicit instruction of comprehension skills, using graphic organizers, and teaching students to make predictions prior to reading. Results added further support for anaphoric cueing and prereading question generation. El Zein et al. also examined methodological quality following Schlosser and Sigafoos's (2007) criteria and concluded that there were few high-quality studies.

Accardo's (2015) review augmented El Zein et al. (2014), identifying 13 group and single-subject research design (SSRD) studies published from 1994 to 2014. Seven of the studies were not included in previous reviews, and the author noted the effective use of systematic prompts in addition to anaphoric cueing, graphic organizers, explicit instruction in comprehension skills, and pre-reading question generation. Accardo also raised concerns about study quality and suggested that due to the quality concerns, strategies identified in the review were of questionable practical use to classroom teachers.

The most recent meta-analytic review, Singh et al. (Singh et al., 2021), focused on SSRD studies, grouping the studies into visually cued, collaborative strategies, metacognitive strategy instruction, technology-assisted instruction, adapted texts, and behavioral strategies to better identify effective practices. They used the What Works Clearinghouse (WWC, 2020) standards to examine study quality and found that visually cued, metacognitive strategy instruction, and adapted texts were the most effective strategies in fostering reading comprehension. They suggested that more effective studies that meet WWC single-case standards without reservations are needed.

In sum, there have been nine systematic reviews inclusive of, or focused on, reading comprehension interventions. These reviews have contributed to a growing general awareness of promising practices. Furthermore, they highlighted the need to examine design quality and methodological rigor in interpreting intervention outcomes. What the reviews to date have not done is relate intervention outcomes to a theory of how readers comprehend text. As a result, although the field has a growing sense of what works, the how or why is not known and consequently researchers cannot systematically extend the known findings in designing high-quality interventions. There is an urgent need to do so given the press for reading achievement by all students expressed in the Every Student Succeeds Act (2015) and the substantial difficulties with reading comprehension exhibited by students with ASD (Huemer & Mann, 2010; Nation et al., 2006; Nation & Norbury, 2005).

The C-I Model

Our research extends the literature in important ways. First, we adopted the C-I model of text comprehension

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(Kintsch, 1988, 1998, 2019) and its classroom applications to guide a more purposeful examination not just of what works but also why. The C-I model proposes that readers engage text by applying construction and integration processes that enable the creation of mental representations of text at surface, micro- and macro-textbase, and situation model levels. Construction begins with *surface*-level processing of linguistic codes (e.g., syllables, word sequence) which involves little semantic processing (Van den Broek & Gustafson, 1999). The semantic level of representation is carried both in sentence-level microstructures, which Kintsch (2019) described as a network of propositions, and discourse-level macrostructures (i.e., the global organization of these ideas). Microstructures, then, are constructed by readers as they determine units of text-based meaning, or propositions, and the associations between them both within and across sentences (Kintsch, 1988; Kintsch & Rawson, 2005). Macrostructures, on the other hand, are constructed as readers recognize semantic relations between larger units of text such as categorizations of information into topics and subtopics. Together, the micro- and macrostructures form the reader's *textbase model* of reading comprehension. During the integration process, the reader integrates relevant background knowledge which has been activated by the textbase to create a coherent mental representation for the text. This process results in creation of a *situation model* (Kintsch, 2019) as readers integrate their schemata with the textbase (Pearson & Cervetti, 2015).

The quality of representing a situation model of the text may be associated with reading ability. Barth et al. (2015) and Denton et al. (2015) found that when compared to average readers, students with reading disabilities struggled with integrating background knowledge to solve comprehension problems, suggesting less difficulty with construction processes and greater challenges with integration. Similarly, for students with ASD, studies found that schemata and background knowledge were activated during their reading, but such activation did not successfully transfer into solving comprehension problems, again suggesting greater challenges with integration (Wahlberg & Magliano, 2004). Researchers attributed such challenges to ToM, EF, and CC deficits (e.g., Carnahan & Williamson, 2010). Previous intervention studies involving typically developing striving readers have demonstrated the effectiveness of integration interventions on reading comprehension (see Elleman, 2017, for a meta-analysis). However, such interventions have rarely been advocated in teaching reading comprehension to students with ASD. Thus, for students with ASD, a richer theoretical perspective in reading comprehension would provide a more purposeful examination of what works and why, potentially providing more systematic and specific guidance in the design and delivery of reading comprehension interventions and research.

The Current Study

The current systematic review and meta-analysis relies on Kintsch's C-I model (Kintsch, 1988, 1998, 2019). This model of the cognitive processes employed by readers as they engage in text comprehension has become the most widely accepted theory of reading comprehension, impacting cognitive psychology and reading curricula, instruction, and assessment (Pearson & Cervetti, 2015). Adopting the C-I model can help teachers understand whether and how reading interventions for students with ASD have been designed to explicitly teach construction and/or integration skills to ensure successful reading comprehension.

The study included two components: a systematic review and a meta-analysis. The systematic review examined the levels (surface, textbase, and situation models) of text processing fostered during reading comprehension instruction based on the C-I model. The systematic review also includes a quality evaluation of the experimental design in each study to provide design suggestions for future experimental studies on reading comprehension instruction for students with ASD. We also used a meta-analysis to quantitively synthesize intervention effects across studies and examined whether intervention effects would be varied by levels of text processing during the instruction. We hypothesized that interventions at the situation model level would be associated with better reading comprehension outcomes than those at surface and textbase model levels, Specifically, we asked the following research questions:

1. What level of text representation do reading comprehension interventions target for students with ASD?
2. What is the effect of reading comprehension interventions on reading comprehension outcomes for students with ASD?
3. What is the quality of reading comprehension intervention studies for students with ASD?

Method

The search process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (Moher et al., 2009) protocol. A comprehensive systematic electronic database search was conducted in seven databases (CINAHL, ERIC, MEDLINE, ProQuest, PsycARTICLES, PsycINFO, and Web of Science) using the terms: (a) *read* literacy* OR *comprehen** OR *text structure* OR *narrative* OR *expository* OR *informational* AND (b) *autis** OR *ASD* OR *asperg** OR *PDD* OR *pervasive developmental disorder** OR *pervasive developmental disabilit** OR *developmental disabilit** OR *developmental delay* OR *hyperlexi**. Reference lists from prior published reviews were also examined.

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This search was conducted in March 2020, and date restrictions were not set.

This review was conducted in three phases. First, the first author screened the titles and abstracts of all articles located by the electronic search to exclude articles that were clearly irrelevant to the review and to remove duplicate records. Also, articles identified by searching reviews were screened. Second, the first and second authors independently conducted a full-text review of all remaining articles to confirm relevant articles that met the inclusion and exclusion criteria. Any disagreements were resolved through consensus. Finally, data from the included articles were extracted by the first and second authors based on the variable definitions provided below and disagreements were resolved through consensus between all three authors.

Inclusion and Exclusion Criteria

Studies met the following criteria were included. First, the study was an empirically based experimental study published in English. SSRD studies that employed a multiple baseline design or a reversal/withdrawal design were included, and group studies that employed a randomized controlled trial or a quasi-experimental design also met this criterion. SSRD studies that used an alternating treatment design (ATD) or group design studies that compared two researcher-designed intervention methods were excluded. This decision was made because the outcome would be a comparison of the effect of one researcher-designed intervention to another, rather than a direct comparison of the effect on reading comprehension as compared to a natural, business-as-usual instruction. Although we did not find group designs that compared two interventions, as shown in Figure 1, we found two ATD studies and they were excluded due to this reason (El Zein et al., 2016; Solis et al., 2016).Second, at least one participant was a kindergarten to grade 12 student with ASD. We relied on the authors' report for the diagnosis of ASD. If a study included students with and without an ASD diagnosis, only outcome data related to students with ASD were analyzed.

Third, the study included a passage-level reading comprehension outcome measure. A study was excluded if there was no outcome measure of passage-level reading comprehension, it did not use or report discourse-level reading comprehension statistics before and after intervention (e.g., measures of word or sentence-level comprehension did not meet this criterion), or if it used symbolated text (i.e., a combination of both text and word-by-word picture symbol representations in alternating lines).

Fourth, reading comprehension assessment was completed independently by students. A study was excluded if students completed reading comprehension assessments with assistance. Studies that had the following forms of assessments were excluded: (1) Receiving help from text-to-speech technologies, (2) answering questions with prompts from teachers, peers, or technologies, or (3) the texts incorporated read-aloud features during the measurement procedures. Text-to-speech technologies did not meet this criterion and were excluded. Text-to-speech technologies were excluded as they would convert the texts into audio presentation and thus require listening comprehension rather than reading comprehension. We did not set decoding ability restrictions as many screened studies did not report decoding (i.e., only 11 out of the 26 included studies reported decoding; see Table 1). We assumed that students with ASD who completed the reading comprehension assessments had sufficient decoding abilities to complete the reading comprehension measure.

Search and Selection Procedure

The electronic search (n = 2695) and reference list review (n = 58) yielded 2753 studies. After de-duplication using Mendeley Reference Manager, 2415 studies were included for title and abstract screening for relevance examination. After reading the title and abstract, 161 studies were included in the full-text screening to determine eligibility for the review.

As shown in Figure 1, 139 studies did not meet the inclusion criteria and were excluded. Studies were excluded because they (1) used a non-experimental design (n = 16) or used an ATD (n = 2), (2) no participants were diagnosed with ASD (n = 57) or the age of participants with ASD outside of the range of kindergarten to grade 12 (n = 5); (3) did not report reading comprehension statistics (n = 46), the outcome measure was not assessing discourse-level reading comprehension (n = 5), or the passage for the outcome measure contains symbolated text (n = 4); and (4) the outcome measure was not completed independently by the students (n = 4).

Data Extraction and Coding

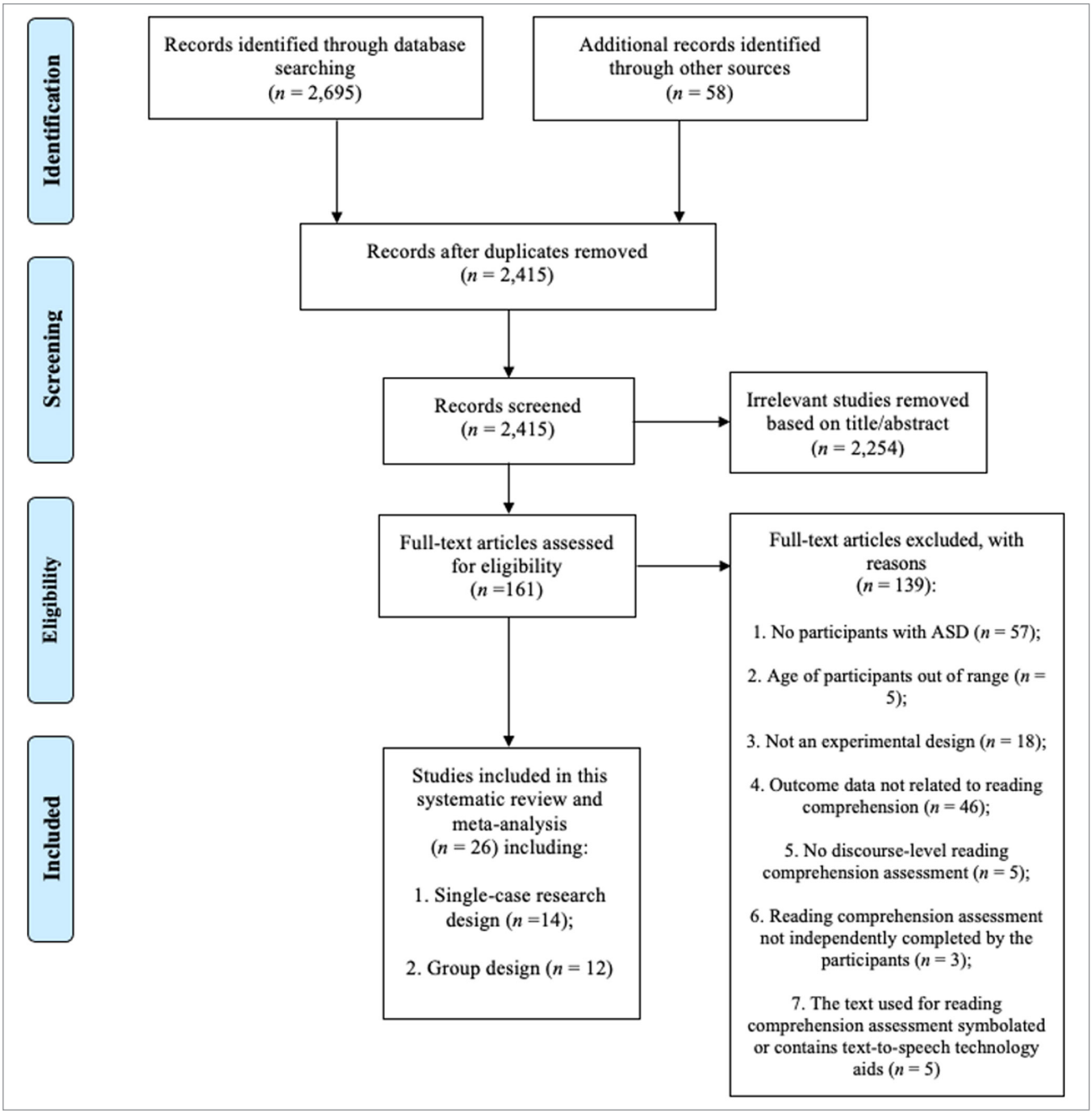
Data were extracted and coded using author agreed-upon definitions of the. Data were coded into the following categories: (a) characteristics of each participant/group including gender, age, specific diagnosis, cognitive profiles, and reading scores; (b) characteristics of the intervention; (c) intervention practices based on the C-I model; (d) study outcomes; and (d) methodological quality and risk of bias.

Characteristics of Children with ASD

Multiple characteristics of the children with ASD were coded: (a) age; (b) gender; (c) specific diagnosis, (d) presence of comorbid conditions; (e) IQ level (full IQ, verbal IQ/language comprehension, and performance IQ); and (f) reading levels. We reported reading levels using norm-referenced scores or diagnostic decoding (word list reading and passage reading accuracy),

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FIGURE 1
Study inclusion decision tree



fluency (words read correctly per minute in list or contextual format), and reading comprehension (i.e., passage reading comprehension question response accuracy) scores.

Reading Comprehension Intervention Coding

The reading comprehension intervention was coded for (a) setting (i.e., school/classroom, clinic, or home), (b) interventionist, (c) intervention practices (see details below), (e) intervention dosage (e.g., individual session duration, frequency, and total intervention duration).

Intervention activities, the core of the intervention components, were coded as surface, textbase (micro and macro level), and situation model levels. The model-specific activities are described below:

Surface Level

As Kintsch (1988) noted, during the construction phase, the linguistic processing of individual words, including the

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TABLE 1
SSRD Participant Information

				Cognitive and Reading profiles						
Study	Age, grade (Gender)	Diagnosis	Comorbid	FIQ	VIQ/ Language abilities	PIQ	Instructional level	Decoding	Fluency	Reading comprehension (Screening test results)
Barnes and Rehfeldt (2013)	12 y, Grade 6 (M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	Parents report: below average	Parents report: below average
	11 y, Grade 6 (M)	Autism	Yes (Tourette, ADHD)	n.a.	n.a.	n.a.	n.a.	n.a.	Parents report: below average	Parents report: below average
	11, Grade 5 (M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	Parents report: below average	Parents report: below average
Chovanes (2018)	16 y, 9th grade (M)	Autism	No	n.a.	n.a.	n.a.	IRI 2 grades below the grade level	n.a.	2 grades below the grade level (unknown measure)	n.a.
Head et al. (2018)	10 y, 5th grade (M)	Autism	No	82 (unknown measure)	n.a.	n.a.	n.a.	WRMT_III Word list on average	n.a.	WRMT_III Passage comprehension"1 SD below the mean
	16 y, 10th grade (M)	Autism	Yes (Mild ID and ADHD)	64 (unknown measure)	n.a.	n.a.	n.a.	WRMT_III Word List 1.5 SD below the mean	n.a.	WRMT_III Passage comprehension: 3 SDs below the mean
Howorth, 2015	9 y, 3rd grade (M)	HF	No	WJ-III > 85	n.a.	n.a.	n.a.	WJ-Word list: on average	DIBELS-Passage reading fluency: 10th percentile	WJ- Passage comprehension: 1 SD below the mean
	9 y, 3rd grade (M)	HF	No	WJ-III > 85	n.a.	n.a.	n.a.	WJ- Word list on average;	DIBELS- Passage reading fluency: 10th–25th percentiles	WJ- Passage comprehension: 3 SDs below the mean
	11 y, 5th grade (M)	HF	No	WJ-III > 85	n.a.	n.a.	n.a.	WJ- Word list on average	DIBELS- Passage reading fluency: 10th–25th percentiles	WJ- Passage comprehension: 1 SD below the mean
	10 y, 4th grade (M)	HF	No	WJ-III > 85	n.a.	n.a.	n.a.	WJ- Word list: 1 SD below the mean	DIBELS-Passage reading fluency: 10th–25th percentiles	WJ- Passage comprehension on average
	11 y, 5th grade (M)	HF	No	WJ-III > 85	n.a.	n.a.	n.a.	WJ- Word list: >2 SDs below the mean	DIBELS- Passage reading fluency: 10th–25th percentiles	WJ- Passage comprehension: >3 SDs below the mean.
	12 y, 6th grade (M)	HF	No	WJ-III > 85	n.a.	n.a.	n.a.	WJ- Word list: on average	DIBELS- Passage reading fluency: 10th–25th percentiles	WJ- Passage comprehension: on average

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TABLE 1
SSRD Participant Information (*continued*)

				Cognitive and Reading profiles						
Study	Age, grade (Gender)	Diagnosis	Comorbid	FIQ	VIQ/ Language abilities	PIQ	Instructional level	Decoding	Fluency	Reading comprehension (Screening test results)
Howorth et al. (2016)	10 y, 5th grade (M)	HF	No	WISC-IV = 98	WISC-IV VIQ standard score: 108	n.a.	n.a.	WJ-III-Decoding: >2 SDs above the mean	n.a.	WJ-III- Passage comprehension: 1 SD above the mean
	11 y, 6th grade (M)	HF	No	WISC-IV =97	WISC-IV VIQ standard score: 92	n.a.	n.a.	WJ-III-Decoding: 1 SD above the mean	n.a.	WJ-III- Passage comprehension: 2 SDs below the mean
	11 y, 5th grade (M)	HF	No	WISC-IV = 111	WISC-IV VIQ standard score: 109	n.a.	n.a.	WJ-III-Decoding: >2 SDs above the mean;	n.a.	WJ-III- Passage comprehension: 2 SDs below the mean
	10 y, 5th grade (M)	HF	No	WISC-IV = 91	WISC-IV VIQ standard score: 100	n.a.	n.a.	WJ-III-Decoding: on average	n.a.	WJ-III- Passage comprehension: 2 SDs below the mean
Kamps et al. (1994)	8 y, 1st grade and 2nd grade combined classroom (M)	Autism	No	101 (unknown measure)	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	8 y, second grade (M)	Autism	No	71 (unknown measure)	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	9 y, 3rd grade (M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Kim et al. (2018)	7 y, grade n.a. (M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	6 y, grade n.a. (M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	8 y, grade n.a.(M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

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TABLE 1
SSRD Participant Information (*continued*)

				Cognitive and Reading profiles						
Study	**Age, grade (Gender)**	**Diagnosis**	**Comorbid**	**FIQ**	**VIQ/ Language abilities**	**PIQ**	**Instructional level**	**Decoding**	**Fluency**	**Reading comprehension (Screening test results)**
Marshall (2017)	15 y, 9th grade (M)	Autism	Yes (Mild ID)	WISC-IV = 71	n.a.	n.a.	IRI 5 grades below the grade level	n.a.	n.a.	WRMT-III- Passage comprehension: 3 SDs below the mean
	17 y, 11th grade (M)	Autism	Yes (ADHD)	WISC-IV = 66	WISC-IV = 57	WISC-IV = 79	IRI 9 grades below the grade level	n.a.	n.a.	WRMT-III- Passage comprehension: >4 SDs below the mean
Regelski Jr. (2016)	Grade 3 or 7 (M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	Grade 3 or 7 (M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
	Grade 3 or 7 (M)	Autism	No	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Reutebuch et al. (2015)	15 y, 9th grade (M)	Autism	No	80–100 (unknown measure)	n.a.	n.a.	IRI 6 grades below the grade;	n.a.	n.a.	WJIII- Passage comprehension: 7 grade levels below the grade
	16 y, 10th grade (M)	Autism	No	80–100 (unknown measure)	n.a.	n.a.	IRI 6 grades below the grade;	n.a.	n.a.	WJIII- Passage comprehension: 9 grade levels below the grade
	17 y, 11th grade (F)	HF	No	80–100 (unknown measure)	n.a.	n.a.	IRI 6 grades below the grade	n.a.	n.a.	WJIII- Passage comprehension: 6 grade levels below grade
Schatz (2017)	14 y, 8th grade (M)	Autism	Yes (SI)	n.a.	n.a.	n.a.	n.a.	WIAT-III-Decoding on average	(a) WIAT-III-Word list fluency-12th percentile; (b) CBM-Passage reading fluency-10th–25th percentiles	(a) WIAT-III Passage comprehension: 5th percentile; (b) CBM-MAZE- Passage comprehension:10th percentile
	13 y, 7th grade (M)	Autism	Yes (SI)	n.a.	n.a.	n.a.	n.a.	WIAT-III-Decoding on average	(a) WIAT-III-Wordlist fluency--on average; (b) CBM-Passage reading fluency-10th–25th percentiles	(a) WIAT-III Passage comprehension: 10th percentile; (b) CBM-MAZE-comprehension: 10th–25th percentile
	12 y, 6th grade (M)	Autism	Yes (SI)	n.a.	Oral language from state test: 12th percentile	n.a.	n.a.	WIAT-III-Decoding on average	(a) WIAT-III-Wordlistt fluency--on average; (b) CBM-Passage reading fluency-10th–25th percentiles	(a) WIAT-III Passage comprehension:14th percentile; (b) CBM-MAZE: Passage comprehension:10th–25th percentiles

(continued)

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TABLE 1
SSRD Participant Information (*continued*)

				Cognitive and Reading profiles						
Study	**Age, grade (Gender)**	**Diagnosis**	**Comorbid**	**FIQ**	**VIQ/ Language abilities**	**PIQ**	**Instructional level**	**Decoding**	**Fluency**	**Reading comprehension (Screening test results)**
Singh et al. (2017)	13 y, 7th grade (M)	HF	No	n.a.	n.a.	n.a.	n.a.	WRMT-III-decoding: 1.75 years above the chronological age;	n.a.	WRMT-Passage comprehension: 1.67years below the chronological age
Solis et al. (2021)	14 y, 8th grade (M)	Autism	Yes (ADD)	n.a.	KBIT-VIQ Standard Score: 60	n.a.	n.a.	WJ-III-Word list-On average	WJ-III- Word list fluency- > 1 SD below the mean	WJ-III- Passage comprehension- > 3 SDs below the mean
	14 y, 8th grade (M)	Autism	Yes (SI; Mild ID)	n.a.	KBIT-VIQ Standard Score: 65	n.a.	n.a.	WJ-III-Word list-On average	WJ-III- Word list fluency-2 SDs below the mean	WJ-III- Passage comprehension- > 4 SDs below the mean
	12, 6th Grade 6 (M)	Autism	Yes (SI; Mild ID)	n.a.	KBIT-VIQ Standard Score: 53	n.a.	n.a.	WJ-III-Word list-On average	WJ-III- Word list fluency-1 SD below the mean	WJ-III- Passage comprehension-1 SD below the mean
	14 y, 8th grade (M)	Autism	Yes (SI; Mild ID)	n.a.	KBIT-VIQ Standard Score: 69	n.a.	n.a.	WJ-III-Word List: >4 SDs below the mean	WJ-III- Word list fluency - > 2 SDs below the mean	WJ-III- Passage comprehension- > 3 SDs below the mean
	14 y, 8th grade (M)	Autism	Yes (Severe ID)	n.a.	KBIT-VIQ Standard Score: 40	n.a.	n.a.	WJ-III-Word list: >3 SDs below the mean	WJ-III- Word list fluency-3 SDs below the mean	WJ-III- Passage comprehension-3 SDs below the mean
Stringfield et al. (2011)	11 y, 5th grade (M)	HF	No	>80 (unknown measure)	n.a.	n.a.	WIAT-II composite reading: 2.4 grades below the grade level	n.a.	n.a.	n.a.
	8 y, 3rd grade (M)	HF	No	>80 (unknown measure)	n.a.	n.a.	WIAT-II composite reading: 1 Grade below the grade level	n.a.	n.a.	n.a.
	8 y, 3rd grade (M)	HF	No	>80 (unknown measure)	n.a.	n.a.	WIAT-II composite reading: >2 grades below the grade level	n.a.	n.a.	n.a.

Note. ADD/ADHD = Attention-deficit/hyperactivity disorder; CBM = Curriculum-based measure; F = female; FIQ = Full IQ; HF = high-functioning autism; ID = intellectual disabilities; IRI = Informal reading inventory; M = male; PIQ = Performance IQ; KBIT = Kaufman Brief Intelligence Test; SI = Speech and/or language impairment; VIQ = Verbal IQ; WIAT = Wechsler Individual Achievement Test; WJ-III = Woodcock Johnson Tests of Achievement-Third Edition; WRMT-R = Woodcock Reading Master Test-Revised; y = years in age; YARC = York Assessment of Reading Comprehension.

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sound, spelling, and semantics and syntactic roles of the words, are activated to lay the foundation for textbase processing. Activities at this level include decoding (e.g., decodable reader practice), vocabulary or word knowledge (e.g., identify the meaning of a keyword or drawing semantic webs of a word), reading fluency practice (e.g., echo and choral reading) that does not involve explicit sentence/passage comprehension instruction, and independent reading for entertainment purposes without teacher scaffolding on sentence/passage comprehension (e.g., reading assigned reading materials based on topic interests). It should be noted that if an activity involves surface level and higher-level (textbase or situation) activities, the intervention was coded at the higher level. For example, Reutebuch et al. (2015) included vocabulary and literal comprehension question-answering activities, and this study was coded at textbase level.

Textbase (Micro and Macro) Level

Micro-level textbase activities involve literal interpretations of individual words or understanding the literal meaning of individual sentences. These activities included defining vocabulary, resolving anaphora, paraphrasing, and answering literal questions (e.g., *who* and *where*).

At a macro-level, students construct information from paragraphs and passages. Questions that involve macro-level textbase processing include recall, summary, and generation of main ideas without teacher scaffolding. We acknowledge that some readers might have integrated background knowledge to answer these questions. Still, without teacher scaffolding, this integration, if any, is implicit and relies on student abilities in the moment rather than teacher instruction.

Situation Model Level

Activities that tapped into a situation model level explicitly addressed background knowledge retrieval and applying knowledge in context. These included teacher-directed question-answering activities that explicitly addressed background knowledge before, during, or after reading. Additionally, if the activities require students to act out the meaning of the texts (e.g., role-playing), those activities were coded as situation model level. Regardless of the labels of the questions (e.g., some authors may have claimed they used inferential questions), if authors did not provide details of the activities to allow confirmation of their description, the level of text representation was coded as the textbase level instead of situation model level.

SSRD Study Outcomes

Tau-U

An examination of the outcome was conducted using a nonparametric measure, Tau-U (Parker et al., 2011). Tau-U is an index of the percentage of data that shows improvement over time, controls for therapeutic baseline trend (if any), and is less influenced by a small number of data points when compared to other methods (Parker et al., 2011). Building on Kendall's distribution, Tau-U provides estimation probability indexes for significance testing and practical significance interpretations and thus is more robust than other effect size measures in SSRD studies (Parker et al., 2011). The formula is:

$$Tau\ U = \frac{S_t - S_b}{mn}$$

In the above formula, S is Kendall's S statistic for baseline-treatment comparison, the sum of the differences between all the paired Mann–Whitney U scores (i.e., $U_{baseline}$–$U_{intervention}$). $S_t - S_b$ is the difference between the S statistics across intervention and control phases. The number of pairs is the product of m (i.e., number of observations in the baseline) and n (i.e., number of observations in the intervention). Kendall's S at baseline is calculated from the paired baseline U scores from which the number of pairs in the baseline phase was calculated as $m^*(m-1)$. The following criteria were applied when interpreting the results: (a) weak or small effects, 0–0.65; (b) medium effects, 0.66–0.92; and (c) large effects, 0.93–1.0 (Parker et al., 2011).

Visual Analysis

Visual analysis was used to examine patterns in the data and to determine if a functional relation (i.e., causal relation) was present (Barton et al., 2018). Procedures for visual analysis include examining changes in level, trend, and variability within conditions and immediacy of change, overlap, and consistency between conditions. A functional relation exists if there are a minimum of three replications of behavior change when the intervention is introduced (Barton et al., 2018).

For this review, visual analysis was conducted using the Systematic Protocols for the Visual Analysis of Single-Case Research Data developed by Wolfe et al. (2019). These protocols, based on published standards and guidelines for SSRDs, provide step-by-step direction for analyzing graphs and generating an experimental control score for each design (Wolfe et al., 2019). The rating scale for the design quality score ranges from 0 to 5, 0–2 indicates a functional relation was not present, 3–5 indicates that a functional relation was present. Moreover, scores in the 3–5 range also indicate the level (i.e., small, medium, large) of behavioral change.

Group Design Study Outcomes

For group design studies, standardized mean differences (SMD) were calculated to measure the gain score (i.e., posttest–pretest) differences between the two groups divided by the pooled standard deviation at pretest. This

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method has less bias than other intervention effect calculation methods (Morris, 2008). We began by calculating Cohen's *d*s to represent SMD, using the formula below:

$$d = \left[\left(M_{t.post} - M_{t.pre}\right) - \left(M_{c.post} - M_{c.pre}\right)\right] / S_{pooled.pre}$$

where,

$$S_{pooled.pre} = \frac{\left(N_{t.pre} - 1\right) * SD^2_{t.pre} + \left(N_{c.pre} - 1\right) * SD^2_{c.pre}}{N_{total.pre} - 2}$$

In the above formula, *t* is the treatment group, *c* is the control group. $S_{\text{pooled.pre}}$ is the pooled *SD* across pretest scores of the intervention and control groups. Due to each study's small sample size, we converted each Cohen's *d* to Hedges's *g* to improve estimation precision. Based on the recommendation by Borenstein et al. (2011), we used a sample correction factor and multiplied it with Cohen's *d* to calculate the Hedges's *g*. Specifically, the procedure is:

$$Correction\ factor = 1 - \frac{3}{4 * \left(N_{total.pre} - 2\right) - 1}$$

$$Hedges'\ s\ g = correction\ factor * d$$

Different from Tau-U that ranges from −1 to 1, *g* does not have a range limit and is interpreted as: (a) small effect (< = 0.20); (b) medium effect (0.21–0.79); and (c) large effect (> = 0.80; Cohen, 1988).

Methodological Quality

We evaluated the studies for both methodological quality and risk of bias. Quality refers to the extent to which a study included components that are widely considered necessary. Risk of bias refers to the likelihood that there is bias within the study that is due to the researcher's decisions in design and implementation that could potentially impact the reported outcome (Ledford et al., 2020).

Two different standards were adopted to evaluate the quality of the included studies. We applied the Single-Case Analysis and Design Framework (SCARF; Ledford et al., 2020) to evaluate rigor and quality of SSRD and What Works Clearinghouse Standards (WWC Version 4.1, 2020) to the group designs. The SCARF was chosen in lieu of WWC standards for SSRD because it incorporates an evaluation of study rigor with visual analysis, a cornerstone for the methodology (Ledford & Gast, 2018). We conducted risk of bias assessments to examine domains of potential systematic errors such as selection bias, performance bias, and detection bias (Higgins & Altman, 2008). We assessed group design studies using the Cochrane Collaborations Tool for Research Bias (Sterne et al., 2019) and assessed SSRD studies using the Single-case Risk of Bias tool (SCD RoB; Reichow et al., 2018).

SCARF for SSRD

The SCARF (Ledford & Gast, 2018) was developed to evaluate SSRD studies, which allows for examination of multiple designs within a single study, if needed. Scores on the SCARF range from 0 to 4 and are displayed on a scatter plot, whereas each design is represented by a single point. The SCARF assesses three quality areas: rigor, quality and generality, and outcome evaluation. Rigor, or internal validity, includes evaluation of three areas: (1) reliability, (2) data sufficiency, and (3) fidelity. Quality and generality include evaluation of the descriptions of the (1) participants, (2) conditions, (3) dependent variables, (4) social and ecological validity and measurement, (5) maintenance, (6) stimulus generalization, and (7) response generalization. We interpreted the scatter plot by separating the graph into four quadrants. Data points that are plotted in the upper right quadrant represent studies with high rigor and positive outcomes (Ledford & Gast, 2018).

WWC Standards for Group Designs

The application of the WWC standards allowed for the examination of group studies on (1) randomization, (2) sample attrition, (3) baseline equivalence if high attrition or quasi-experimental, and (4) non-design components (e.g., psychometric properties, confounding factors). Based on the WWC standards (WWC, 2020), each group experimental study was assigned a rating of either *meets without reservations*, *meets with reservations*, or *does not meet standards*.

Risk of Bias

For group experimental designs, the Cochrane Collaboration's Tool for Research Bias (Sterne et al., 2019) was used to assess five sources of potential bias. Each source of bias was rated as *low*, *some concerns*, or *high risk of bias* based on the algorithm provided. The five domains that were assessed could be biased due to (1) randomization process, (2) effects of assignment and adherence to the intervention, (3) missing outcome data, (4) measurement of outcome, or (5) reported results. An overall judgment of the study was assigned based on the following criteria: (a) overall low risk of bias if all domains were judged to be low risk, (b) some concerns if at least one of the domains were judged with some concerns, but no domains were high risk, and (c) high risk if at least one domain was judged to be high risk. The single case design risk of bias tool (SCD RoB; Reichow et al., 2018) served to modify the evaluation of bias described above for application to SSRDs. Seven domains of bias were evaluated, including (1) sequence generation, (2) participant selection, (3) blinding of participants and personnel, (4) procedural fidelity, (5) blinding of outcome assessment, (6) dependent variable reliability, and (7) data sampling. An overall judgment of the study bias was assigned based on the following criteria: (a) low risk, (b) unclear risk, or (c) high risk.

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Interobserver Agreement

Interobserver agreement (IOA) was determined at multiple times throughout the review process. First, during initial full-text screening, all 161 studies were reviewed for inclusion by the first and second authors. IOA was calculated by the formula: agreements/(agreements + disagreements) ×100. For screening, agreement was 92.5%. Then IOA was calculated on quality ratings for group and SSRD designs. There was 100% agreement on the WWC rating for group studies. For the SCARF, mean agreement across all studies was 83.7% (*SD* = 5.1, Range = 74.5%–91.5%).

Meta-Analytic Procedures

Because reading comprehension was measured variably across studies (e.g., composite assessment vs. retelling only), we calculated each Tau-U and *g* at the measure level. We used the random effects robust standard error estimation technique to address sample dependence. The R package *robumeta* (Fisher & Tipton, 2015) was used for this estimation. Using this package, we set the between-measure correlation as 0.80 and then assessed the between-study variance (τ^2). We also used the *Q* statistic and its associated *p* values to determine the significance of τ^2, and the I^2 estimates the ratio of τ^2 to the total variance (Borenstein et al., 2011). Forest plots from the intercept models were graphed to reflect individual study effects and the overall effect.

Moderator Analysis

Our primary interest was how the instructional focus on the readers' level of text representation (i.e., surface, textbase, and situation model levels) would influence meta-analysis outcomes after controlling for the type of measure and instructional dosage. Types of measures included curriculum-based measures (CBMs) of composite comprehension, retelling, cloze, and norm-referenced measures. CBM composite assessments included *wh*- questions and were assessed by the percentage of questions answered correctly; retelling assessed students' main idea recall abilities and was determined by the number of key ideas recalled (e.g., Howorth, 2015) or the number of keywords recalled on a rubric (e.g., Marshall, 2017). Cloze measures included filling in missing information via multiple choice or oral cloze tasks. Norm-referenced measures of reading comprehension only appeared in group design studies, and we separated norm-referenced measures and curriculum-based composite measures. Instructional dosage was treated as a continuous variable, and the total minutes per participant/group served as a proxy for this variable. Some studies gave a range of minutes, and for those studies, we used the midpoint dosage ([min + max minute]/2) to represent the average dosage. Both *instructional level of text representation* and *dosage* were coded at the study level, and *measure* was coded at the effect size level.

Publication Bias

We applied the (1) three parameter-selection model (3PSM) and (2) sensitivity analysis with the robust standard error estimation to assess publication bias. The 3PSM examines the likelihood that studies with significant results (i.e., $p < 0.05$) have a better chance of being published than those with nonsignificant results ($p > 0.05$). The 3PSM has demonstrated fewer chances for Type-I errors to occur than traditional methods such as Egger's regression test or Trim-and-Fill analysis (Pustejovsky & Rodgers, 2019). We additionally provided a sensitivity analysis of publication bias by assessing would the synthesized effect size still be significant if only nonsignificant group studies were published. The 3PSM and sensitivity analysis were conducted, respectively, in R *weightr* (Coburn & Vevea, 2019) and *PublicationBias* (Mathur & VanderWeele, 2019) packages.

Results

Diagnosis, Cognitive, and Reading Profiles

SSRD

Table 1 shows the participant information for the SSRD studies. The 14 included SSRD studies had a total of 42 students (1 female) that met inclusion criteria. These students ranged in age from 6 to 17 years. In all, 15 of the students (36%) were described as having high-functioning autism. Four students with ASD (10%) had comorbid attention-deficit/hyperactivity disorder (ADD/ADHD), and one of these four additionally had Tourette syndrome. Four students with ASD (10%) had speech and/or language impairments. For diagnosis criteria, 18 of the 26 studies (69%) reported that inclusion for the study relied on a clinical diagnosis of ASD. However, not all specified what assessment (e.g., ADOS) was used for the diagnosis and it was more common for studies to report that there was a clinical/medical diagnosis and not report the assessment. On the other hand, 8/26 (30%) used an IDEA educational classification of ASD or a parent confirmation of a medical diagnosis.

Where reported, students' decoding ranged from four *SD*s below the average (Solis et al., 2021) to 2 *SD*s above the average (Howorth et al., 2016). A total of 29% students had decoding scores at least one *SD* below the mean. All students had fluency scores at least one *SD* below the mean. All but three students showed reading comprehension challenges (= > 1 *SD* below the mean) at the screening test. In all, 15 studies (58%) had no decoding reports, 20 studies (77%) studies had no fluency reports, and 12 (46%) studies had no reading comprehension reports at screening stages.

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Group Design

Table 2 shows the participant information for the group studies. The 12 group design studies had 323 participants, including 167 from the treatment group and 156 from the control group (16% and 13% female students from the treatment group and control group, respectively). Only Bailey et al. (2017) and Turner et al. (2017) reported comorbid conditions (ADHD) with ASD. All of the group studies reported norm-referenced language comprehension or verbal IQ test scores, and six studies (50%) had students with Verbal IQ scores below 85 (below average). Six studies reported Performance IQ scores, and two studies (Turner et al., 2017; Uccheddu et al., 2019) had participants with Performance IQ below 85. Among the studies that adopted norm-referenced reading measures, three studies reported decoding mean standard scores above 85, and four studies reported mean fluency standard scores above 85. For studies that reported reading comprehension standard scores, Turner et al.'s (2017) study was the only study that had a mean standard reading comprehension score above 85.

Characteristics of the Intervention

Table 3 shows the design information for each SSRD and group study. Children with ASD were typically receiving intervention in a school setting (n = 17). Other interventions were conducted in clinics (n = 4) or homes (n = 4). Despite the large number of studies conducted in schools, interventions were conducted predominantly by researchers (n = 23). Three studies were conducted by both a researcher and a teacher (i.e., teacher was trained to implement the intervention) and only two studies were conducted solely by the classroom teacher. One intervention was conducted by a psychologist. On average, a reading intervention was implemented 3.1 sessions per week (SD = 1.4; range = 1 to 5) with an average session duration of 64.1 min (SD = 72.9; range 15 to 240). Notably, three studies were implemented in a clinic for 4 h per session, which influenced the average session duration data as all other studies had at most 70 min per session. Without these studies, the average session duration was 36.3 min (SD = 15.1; range = 15–82.5). The total dosage across studies ranged from 40 min to 12,000 min, and the median dosage was 675 min.

As also shown in Table 3, three studies (2 SSRD and 1 group design) focused on surface-level skill instruction, 15 (8 SSRD and 7 group designs) focused on textbase-level instruction, and eight (4 SSRD and 4 group designs) addressed a situation model level of instruction.

Outcome

Figure 2 shows the effect sizes in forest plots. Tables 4 and 5 show the overall effects and moderator analysis results.

Tau-U and Visual Analysis

As shown in Table 4, the Tau-U for immediate effect was small (Tau-U = 0.62, SE = 0.08, p < .001, 95% CI = [0.44, 0.80]) and for the maintenance effect was medium (Tau-U = 0.81, SE = 0.10, 95% CI = [0.66, 1.00]). Of the multiple-baseline designs (n = 12), five studies (41.7%) demonstrated a functional relation between the intervention and the reading comprehension measure. These five studies demonstrated strong to medium effects of the reading comprehension intervention with scores on the Visual Analysis Protocol (Wolfe et al., 2019) ranging from 4.56 to 4.78.

Hedges's g

As shown in Table 5, a small yet significant effect on reading comprehension was found for group designs (g = 0.26, SE = 0.06, p < .001, 95% CI = [0.13, 0.39]). Immediate post-tests generated a larger effect size value (g = 0.29, SE = 0.07, p < .001, 95% CI = [0.13, 0.46]) than the two follow-up post-tests from maintenance phases (g = 0.15, SE = 0.04, p < .001, 95% CI = [0.07, 0.24]).

Publication Bias

For SSRD studies, the 3PSM likelihood test did not find the possibility that statistically significant findings were more likely published than nonsignificant findings ($\chi 2$ = 0.69, p = .41 > .05). Also, the sensitivity analysis suggested that even after removing all the significant results, the remaining nonsignificant findings could still demonstrate a much reduced yet significant overall effect (Tau-U = 0.20, SE = 0.06, p < .05, 95% CI = [0.03, 0.38]). Only one group study (Engel & Ehri, 2020) had a significant Hedges's g statistic (from the retelling measure), and thus the 3PSM likelihood test did not find the possibility that significant findings were more likely being published than nonsignificant findings ($\chi 2$ = 0.71, p = .40 > .05). Also, even after removing Engel and Ehri's (2020) retelling score, the overall effect was still significant (g = 0.24, SE = 0.05, p = .003 < .01, 95% CI = [0.13 0.36]). In conclusion, we did not detect publication bias threats.

Moderator Analysis

We first conducted a subgroup analysis to investigate how reading comprehension performance differed by level of text processing. Overall, the subgroup analysis suggested advantages of a situation model level of instruction

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TABLE 2
Group Study Participant Information

Study	Design	Age (SD)	Researcher-reported diagnosis	Comorbidity	Full IQ	Verbal IQ/language comprehension	Performance IQ	Decoding standard scores	Fluency standard scores	Reading comprehension pretest/screening standard scores
Arciuli and Bailey (2019)	RCT (*n* = 23 T-11M 0F; C-12M 0F)	T: 7.34 y (SD n.a.); C: 7.34 y (SD n.a.)	Autism	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Bailey et al. (2017)	RCT (*n* = 19 T-8M 1F; C-10M 1F)	T: 7.27 y (1.64); C: 7.52 y (1.56);	Autism	T- 4 ADHD; C-10 ADHD	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Campbell (2010)	RCT (*n* = 10; T-5M 0F; C-5M 0F)	7–12 y (group information n.a.)	Autism	n.a.	n.a.	(1) TOLD-III-Grammar Scaled score (*M* = 10; *SD* = 3): T-5.20 (7.20); C-2.77 (2.04); g = 0.44 (Both groups below average) (2) TOLD-III Listening comprehension; T: 61.00 (7.65) C: 62.60 (12.54) (Both groups below average)	n.a.	WJ III Word List: T: 69.00 (20.58); C: 77.50 (9.54) g = −0.51 (Both groups below average)	n.a.	WJ-III- Passage Comprehension: T-53.80 (11.34) C-51.60 (9.18); g = 0.21 (Both groups below average)
Engel and Ehri (2020)	RCT (*n* = 20; T- 10M 0F C- 8M 2F)	T-7.06 y (.92) C-7.07 y (.69);	HF	n.a.	n.a.	PPVT-4: T-100.60 (5.74) C-99.10 (8.05); g = 0.21; (Both groups on average)	n.a.	n.a.	n.a.	n.a.
Kamps et al. (2016)	QED (*n* = 62; T-29M 3F C-24M 6F)	T: 67.9 months (4.7) C: 68.9 months (5.0)	Autism	n.a.	n.a.	PPVT-4: T-81.1 (14.5); C-79.9 (17.4); g = 0.08 (Both groups below average)	n.a.	n.a.	n.a.	n.a.

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TABLE 2
Group Study Participant Information (*continued*)

Study	Design	Age (SD)	Researcher-reported diagnosis	Comorbidity	Full IQ	Verbal IQ/language comprehension	Performance IQ	Decoding standard scores	Fluency standard scores	Reading comprehension pretest/screening standard scores
Lemelman (2016)	QED (*n* = 39; T- 18M 4F C- 14M 3F)	T: 10.29 y (1.53) C: 11.05 y (1.15);	HF	n.a.	WASI-FIQ T-93.00 (13.31); C- 96.06 (14.91); g = −0.22	WASI-VIQ: T- 90.29 (10.64) C- 90.71 (12.96); g = −0.05; (Both groups on average) PPVT: T-90.68 (11.48) C-90.59 (12.53); g = −0.01 (Both groups on average)	WASI-PIQ T-100.16 (19.54) C- 102.82 (16.18); (Both groups on average)	(a) SORT-III; T-108.14 (8.14); C-105.71 (8.15); (Both groups on average) (b) GORT-passage accuracy: T-98.64 (8.89); C-90.88 (12.28); (Both groups on average)	n.a.	GORT-4- Passage Comprehension: T-77.5 (11.92) C-83.53 (12.96); g = −0.49 (Both groups below average)
Maximo et al. (2017)	RCT (*n* = 28; T-12M 2F; C-12M 2F)	T- 10.0 y (1.2); C- 11.0 y (1.6);	Autism	n.a.	WASI-FIQ: T-94.0 (13.8); C-94.0 (15.8)	WASI-VIQ: T-90.8 (9.8); C-91.9 (13.3); (Both groups on average)	WASI -PIQ: T-103.4 (19.0); C-102.3 (13.3); (Both groups on average)	SORT-R: T-107.6 (8.8) C-106.8 (7.4); (Both groups on average)	n.a.	GORT-4- Passage Comprehension: T-78.0 (13.1) C-85.4 (11.7) (T group below average; C group on average)
Murdaugh et al. (2017)	RCT (*n* = 25; T-13M 1F C-8M 3F)	T: 10.3 y (1.3); C: 11.0 y (1.1);	Autism	n.a.	WASI-FIQ T-90.4 (12.0); C-93.5 (8.3);	WASI-VIQ T-91.5 (10.1); C-88.2 (7.5); (Both groups on average)	n.a.	SORT-R: T-104.8 (5.4); C-107.1 (7.2); (Both groups on average)	n.a.	GORT-4- Passage Comprehension: T-77.5 (12.4) C-84.5 (12.7) (Both groups below average)
Roux, Dion and Barrette (2015)	RCT (*n* = 13; T- 6 C- 7; gender unknown)	T: 8.9 y (1.3); C: 9.2 y (0.8)	Autism	n.a.	n.a.	(a) WISC-IV-verbal comprehension: T-73.3 (10.9); C-68.7 (18.4); (Both groups below average) (b) PPVT-4-age equivalent: T- 7.8 (2.2); C-7.5 (3.2) (Both groups below average)	WISC-IV-Nonverbal reasoning: T-87.2 (23.5) C-96.4 (18.1); (Both groups on average)	n.a.	n.a.	n.a.

(continued)

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TABLE 2
Group Study Participant Information (*continued*)

Study	Design	Age (SD)	Researcher-reported diagnosis	Comorbidity	Full IQ	Verbal IQ/language comprehension	Performance IQ	Decoding standard scores	Fluency standard scores	Reading comprehension pretest/screening standard scores
Roux, Dion, Barrette, Dupéré and Fuchs (2015)	RCT (*n* = 45; T-17M 7F C-20M 1F)	T: 8.8 y (1.1); C: 9.3 y (1.6);	Autism	n.a.	n.a.	(a) WISC-IV-verbal comprehension T-81.0 (14.6) C-79.0 (14.6); g = 0.14 (Both groups below average) (b) PPVT-4-age equivalent: T-7.5 (2.1) C-7.2 (2.5); g = 0.13 (Both groups below average)	WISC-IV-Nonverbal reasoning: T-92.4 (15.3); C-87.5 (17.4); g = 0.30 (Both groups on average)	n.a.	n.a.	n.a.
Turner et al. (2017)	RCT (*n* = 29; T-8M 7F; C-12M 2F)	T-13.63 y (1.19); C-13.27 y (1.30)	Autism	ADHD (sample size n.a.)	n.a.	WISC-IV scale score: T-7.53 (1.96); C-7.50 (2.07); (Both groups on average)	WASI-II-PIQ T-42.40 (12.67); C-42.80 (15.15) (Both groups below average)	BAS-III: T:81.0 (5.33); C: 86.21 (10.74); (T below average; C on average)	YARC-Fluency: T- 89.47 (12.92); C-89.43 (11.90); (Both groups on average)	YARC- Passage Comprehension: T: 88.67 (11.44); C: 93.64 (12.57); (Both groups on average)
Uccheddu et al. (2019)	RCT (*n* = 9; T- 4M 1F C-3M 1F)	T: 7.60 (2.30); C: 8.25 (1.73);	Autism	n.a.	WASI-FIQ T- 75.2 (16.64); C- 108.00 (24.70) (T below average; C on average)	WASI-VIQ: T- 84.7 (5.6) C-102.0 (26.9) (T below average; C on average)	WASI-PIQ-T:79.0 (13.34) C:106.0 (15.7); (T below average; C on average)	n.a.	n.a.	n.a.

Note. ADD/ADHD = Attention-deficit/hyperactivity disorder; C = Control; DIBELS = Dynamic Indicators of Basic Early Literacy Skills; F = female; FIQ = Full IQ; HF = high-functioning autism; M = male; KBIT = Kaufman Brief Intelligence Test; PIQ = Performance IQ; QED = Quasi-experimental design; RCT = Randomized controlled trial; SI = speech and/or language impairment; VIQ = Verbal IQ; T = treatment; WJ-III = Woodcock Johnson Tests of Achievement-Third Edition; WRMT-R = Woodcock Reading Mastery Test-Revised; WASI = Wechsler Abbreviated Scale of Intelligence; y = years; y = year; YARC = York Assessment of Reading Comprehension; Decoding, fluency, and reading comprehension levels are determined based on mean standard scores (*M* = 100; *SD* = 15; on average = 85–115; below average < 85; above average > 115), grade-equivalent scores (on average = on grade level; below average < grade level; above average > grade level), or scaled scores (*M* = 10; *SD* = 3; on average = 7–13; below average <7; above average > 13).

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TABLE 3
Study Design and Instruction Information

Study	Research design, setting, agent	Intervention activities and the C-I model	Intervention dosage: Week total, session frequency per week, minute per session, total cumulative dosage for each participant	Reading comprehension measure
SSRD				
Barnes and Rehfeldt (2013)	• MBL • University lab • Researcher	Teacher models fluent reading, provide corrective feedback, and lead re-reading (Surface)	• Total 4 weeks • 2–3 sessions per week • 30–70 min per session • Total 240–840 min	1. Fill in the blank by answering multiple-choice questions (Cloze) 2. Percent of correct responses to reading comprehension questions (Composite)
Chovanes (2018)	• MBL • School • Researcher	Teacher teaches students to decode multisyllabic words, shrink paragraphs and identify the main idea, eliminate irrelevant information and summarize texts, use signaling words to identify text structure, and answer questions (Textbase)	• Total 4 weeks • 2–3 sessions per week • 30–70 min per session • Total 960–1440 min	Number of items retold correctly on the retell assessment rubric (Retelling)
Head et al. (2018)	• MBL, MP • School • Researcher	A comprehension literacy program containing (a) reasoning, (b) information processing, (c) vocabulary , (d) grammar, (e) basic comprehension, and (f) writing skills (Textbase)	• Total 6 weeks • 10 sessions per week • 30 min per session • Total 1800 min	Percent of correct responses for reading comprehension probes (Composite)
Howorth (2015)	• MBL • School • Researcher	(1) Before reading: Teacher asks questions to activate background knowledge and identify the author's purpose; (2) During reading: Teacher asks inferential questions that address both the textual information and the background knowledge and monitor the fluency and comprehension; (3) After reading: Teacher asks main idea, summary, and inferential questions (Situation)	• Total 9 weeks • 3–5 sessions per week • 45–60 min per session • Total 1215–2700 min	1. Rubric scoring of an oral retell summary statement (Retelling) 2. Percentage of comprehension questions answered correctly (Composite)

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TABLE 3
Study Design and Instruction Information (*continued*)

Study	Research design, setting, agent	Intervention activities and the C-I model	Intervention dosage: Week total, session frequency per week, minute per session, total cumulative dosage for each participant	Reading comprehension measure
Howorth (2015)	• MBL • School • Researcher	(1) Before reading: Teacher asks questions to activate background knowledge and students identify the author's purpose; (2) During reading: Teacher asks inferential questions that address both the textual information and the background knowledge and monitor the fluency and comprehension; (3) After reading: Teacher asks main idea, summary, and inferential questions (Situation)	• Weeks unknown, total of 9 sessions • Weekly frequency unknown • 45 min per session • Total 405 min	1. Rubric scoring of an oral retell summary statement (Retelling) 2. Percentage of comprehension questions answered correctly (Composite)
Kamps et al. (1994)	• Reversal design embedded in MBL • School • Researcher/Peer	Peer provides corrective feedback to fluency errors and ask wh-questions to each other (Textbase)	• Weeks unknown; total 15–35 sessions • 3–4 sessions a week • 25–30 min per session • Total 375–1050 min	Percentage of comprehension questions answered correctly (Composite)
Kim et al. (2018)	• MBL • School • Researcher	(1) Before reading: Teacher activates schema by presenting pictures that are relevant to the text content; (2) During reading: Teacher monitors comprehension by having the focal students discuss with peers about the content; (3) After reading: Teacher asks students retell to summarize and answer summary questions (Situation)	• Total 2 weeks, 6 sessions • 3 sessions week (one session per day) • 25-40-min per session • Total 150–240 min	Percentage of comprehension questions answered correctly (Composite)
Marshall (2017)	• Alternating treatment embedded in MBL • School • Researcher	Matching interest to texts and embed questions of interests into measurements (Surface)	• Total 10 weeks, 8 sessions • Session frequency per week unknown • 5–10 min per session • Total 40–80 min	(1) Oral retelling based on number of words relevant to the main idea (retelling) (2) Percentage corrects on composite questions (composite)
Regelski Jr. (2016)	• MBL • Home • Researcher/Peer	(1) Partner reading aloud, (2) Partner Retelling, (3) Paragraph summarizing (shrinking), (4) Partner prediction (Textbase)	• Total 10 weeks, 8 sessions • Session frequency per week unknown • 5–10 min per session • Total 40–80 min	Fill in the blank by answering multiple choice questions (Cloze)

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TABLE 3
Study Design and Instruction Information (*continued*)

Study	Research design, setting, agent	Intervention activities and the C-I model	Intervention dosage: Week total, session frequency per week, minute per session, total cumulative dosage for each participant	Reading comprehension measure
Reutebuch et al. (2015)	• MBL • Home • Researcher	(1) Before reading: Teacher introduces the topic and identifies key vocabulary word; (2) During reading: Students answer true/false questions and receive feedback; (3) After reading: Students design questions and summarize the text (Textbase)	• Total 16 Weeks • 1 session per week • 20–30 min per session • Total 320–480 min	Percent correct on main idea, literal, vocabulary, inferential, and short answer of questions (Composite)
Schatz (2017)	• MBL • Home • Researcher	(1) Before reading: Students making predictions by previewing the text; (2) During reading: Students Enacting the text by acting out using reader's theater scripts and discuss content with teachers; (3) After reading: Students answer quiz question containing retelling, making predictions to future plots, explicit, and inferential questions (Situation)	• Total 4 weeks • 3 sessions per week • 30–40 min per session • Total 360–480 min	(1) Fill in the blank by answering multiple choice questions (Cloze) (2) Percentage corrects on the Comprehension Questions Protocol (Composite)
Singh et al. (2017)	• MBL • Home • Researcher	(1) During reading: Students answer prediction, summarization, and clarifying question; (2) After reading: Students answer summarization questions (Textbase)	• Total 6 weeks • 3–4 session per week • 30 min per session • Total 540–720 min	Percentage corrects on prediction, summarization, clarification, and summarization questions (Composite)
Solis et al. (2021)	• Alternating treatment embedded in MBL • School • Researcher	(1) Before reading: Teacher selects texts based on interest, previews vocabulary with visuals and student-friendly definition; (2) During reading: Students practice answering literal questions (3) After reading: Students summarize the text and answer literal and summarization questions (Textbase)	• Weeks unknown; total 40 sessions • 5 sessions a week • 30 min per session • Total 1200 min	Number of correct answers to reading comprehension questions (Composite)

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TABLE 3
Study Design and Instruction Information (*continued*)

Study	Research design, setting, agent	Intervention activities and the C-I model	Intervention dosage: Week total, session frequency per week, minute per session, total cumulative dosage for each participant	Reading comprehension measure
Stringfield et al. (2011)	• MBL • School • Researcher	(1) Graphic organizer instruction to identify text structures; (2) Students choose to or not to use graphic organizers after reading; teachers do not provide feedback in either condition. (3) After reading: Students answer composite questions (Textbase)	• Weeks unknown; total 13–22 sessions excluding absence • 5 sessions a week • 15 min per session • Total 195–330 min	Percentage corrects on composite questions (Composite)
Group design				
Arciuli and Bailey (2019)	• RCT • School • Teacher	Teacher asks prediction, sequencing, summarizing, and other comprehension questions for comprehension monitoring (Textbase)	• Total 9 weeks; • 2 sessions per week • 75–90 min per session • Total 1350–1620 min	NARA-3 (Composite)
Bailey et al. (2017)	• RCT • School • Researcher	Teacher asks prediction, sequencing, summarizing, and other comprehension questions for comprehension monitoring (Textbase)	• Total 13 weeks • 2 sessions per week (26 total) • 60 min per session • Total 3120 min	NARA-3 (Composite)
Campbell (2010)	• RCT • School • Researcher	Explicit anaphora instruction followed by comprehension questions (Textbase)	• Total 10 weeks • 2 sessions per week • 20 min per session • Total 400 min	WJ-III- Passage Comprehension (Composite)
Engel and Ehri (2020)	• RCT • School • Researcher	Explicit instructions on ambiguous words, sentences, homographs, anaphora, story sequencing, retelling, and inferencing. (Textbase)	• Total 5 weeks • 6 sessions in total • 30 min per session • Total of 900 min	(1) WRMT-R (Composite) (2) Retelling from Fountas and Pinnell reading passages (Retelling)
Kamps et al. (2016)	• QED • School • Teacher	Teacher teaches phonemic awareness, letter–sound correspondence, decoding, and beginning comprehension skills (Textbase)	• Weeks unknown; Total 60 sessions in Kindergarten, 69 session in First Grade • 3 sessions per week • 30 min per session • Total 1800–2070 min	WRMT-Passage (Composite)

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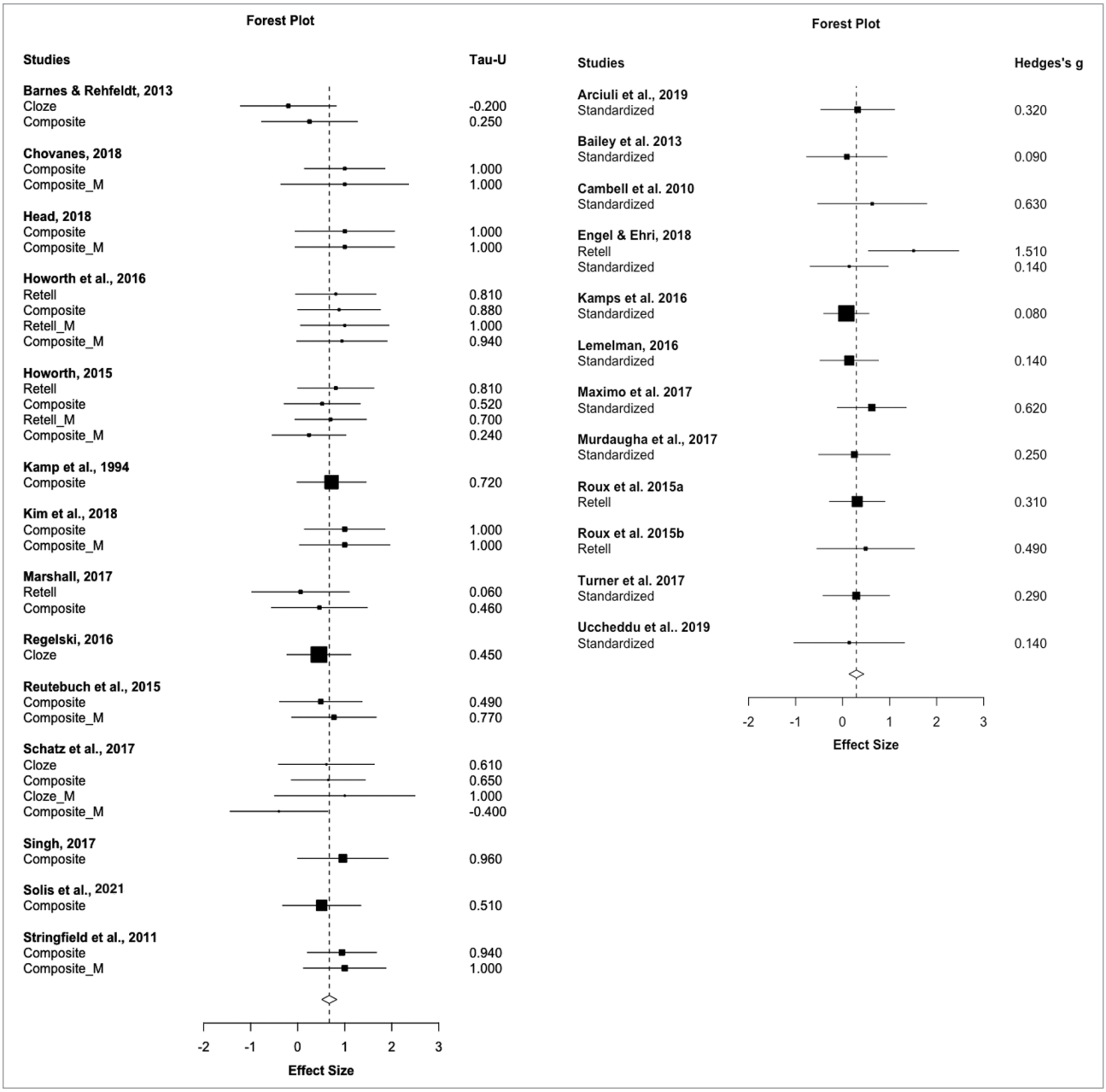
TABLE 3
Study Design and Instruction Information (*continued*)

Study	Research design, setting, agent	Intervention activities and the C-I model	Intervention dosage: Week total, session frequency per week, minute per session, total cumulative dosage for each participant	Reading comprehension measure
Lemelman (2016)	• QED • Clinic • Researcher	Teacher directs students to visualize words, sentence meanings, and helped the students visualize the connection between ideas (Situation)	• Total 10 weeks • 5 sessions per week • 4 h per session • Total 12,000 min	GORT-4- Passage Comprehension (Composite)
Maximo et al. (2017)	• RCT • Clinic • Researcher	Teacher directs students to visualize words, sentence meanings, and helped the students visualize the connection between ideas (Situation)	• Total 10 weeks • 5 sessions per week • 4 h per session • Total 12,000 min	GORT-4- Passage Comprehension (Composite)
Murdaugh et al. (2017)	• RCT • Clinic • Researcher	Teacher directs students to visualize words, sentence meanings, and helped the students visualize the connection between ideas (Situation)	• Total 10 weeks • 5 sessions per week • 4 h per session • Total 12,000 min	GORT-4- Passage Comprehension (Composite)
Roux, Dion and Barrette (2015)	• RCT • School • Researcher	Teacher provides vocabulary, anaphora, main idea, and text structure instruction (no details provided; Textbase)	• Total 16 weeks • 3 sessions per week • 30 min per session • Total 1440 min	Comprehensive questions from the texts (Composite)
Roux, Dion, Barrette, Dupéré and Fuchs (2015)	• RCT • School • Researcher	Teacher provides vocabulary, anaphora, main idea, and text structure instruction (no details provided; Textbase)	• Total 10 weeks (total 15 h) • 3 sessions per week • 30 min per session • Total 900 min	Comprehensive questions from the texts (Composite)
Turner et al. (2017)	• RCT • School • Researcher	Peer tutoring on prediction, summarization, clarifying and teacher directed the peers to ask, probe and answer inferential questions (Situation)	• Total 6 weeks • 2 sessions per week • 45 min per session • Total 540 min	YARC- Passage Comprehension (Composite)
Uccheddu et al. (2019)	• RCT • Clinic • Psychologist	Students read to therapy dogs (Surface)	• 10 weeks • 1 session per week • 30 min per week • Total 300 min	TORC- Passage Comprehension (Composite)

Note. GORT-4 = Gray Oral Reading Test–Fourth Edition; NARA-3 = Neale Analysis of Reading Ability (NARA-3); MBL = multiple baseline design; MP = Multiple probes; RCT = Randomized controlled trial; QED = Quasi-experimental design; TORC = Test of Reading Comprehension (Italian; Valutazione Della Comprensione del Testo Orale); WRMT-R = Woodcock Reading Mastery Test-Revised; WJ-III = Woodcock Johnson Tests of Achievement-Third Edition; YARC = York Assessment of Reading Comprehension.

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FIGURE 2
Forest Pots for Single-Case (Left) and Group Studies (Right).



Note. CBM = Curriculum-based measures; Norm = Norm-referenced tests; _M = Measures for Maintenance phases.

(Tau-Us: Situation = 0.74, Textbase = 0.72, Surface = 0.14; Hedges's *g*s: Situation = 0.31, Textbase = 0.25, Surface = 0.14). The heterogeneity analysis suggested that neither SSRD nor group design studies showed significant *Q* statistics, and the τ^2 and I^2 statistics were null across designs (see the Table 3 note), suggesting there was no significant between-study variance. However, because the heterogeneity statistics are influenced by small sample sizes (Borenstein et al., 2011; von Hippel, 2015), we still performed moderator analyses. As shown in Tables 4 and 5, assessment type explained significant variance of intervention outcomes for SSRD and group design studies, after controlling for dosage and instructional models. In SSRD studies, cloze scores were significantly lower than CBM composite reading scores (cloze vs. composite $b = -0.31$, $SE = 0.12$, $p = .03 < .05$, 95% CI = [−0.58, −0.05]). In group studies, curriculum-based retelling measures had better outcomes than norm-referenced reading measures (retelling vs. composite $b = 1.34$, $SE = 0.06$, $p < .001$, 95% CI = [1.12, 1.46]).

As for levels of text representation, in SSRD studies, we found that surface level showed significantly lower effects

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TABLE 4
Tau-U, Visual Analysis, Estimated Total, and Moderator Analysis

Study	Tau-U	Var.	Sig.	95% CI	Score	FR	Q/R	Out
Immediate intervention								
Barnes and Rehfeldt (2013) (cloze)	−0.20	0.27		[−0.72, 0.32]	-	-	-	-
Barnes and Rehfeldt (2013) (composite)	0.25	0.27		[−0.27, 0.77]	-	-	-	-
Chovanes (2018)	1.00	0.19	***	[0.62, 1.00]	4.56	Yes	2.70	4
Head et al. (2018)	1.00	0.29	***	[0.65, 1.00]	P1: 4.78 P3: 4.56	Yes	P1: 1.47 P3: 1.03	P1: 4 P3: 3
Howorth (2015) (Retell)	0.81	0.17	***	[0.49, 1.00]	1.67	No	2.25	1
Howorth (2015) (Composite)	0.52	0.17	***	[0.19, 0.86]	0.33	No	2.25	1
Howorth et al. (2016) (Retell)	0.81	0.19	***	[0.43, 1.00]	0.36	No	1.78	3
Howorth et al. (2016) (Composite)	0.88	0.20	***	[0.50, 1.00]	1.66	No	1.78	1
Kamps et al. (1994)	0.72	0.14	***	[0.45, 0.99]	1.67	No	1.58	3
Kim et al. (2018)	1.00	0.19	***	[0.62, 1.00]	4.78	Yes	2.92	4
Marshall, 2017 (Retell)	0.06	0.28		[−0.49, 0.60]	-	-	-	-
Marshall (2017) (Composite)	0.46	0.27		[−0.07, 0.99]	-	-	-	-
Regelski Jr. (2016)	0.45	0.12	***	[0.21, 0.70]	-	-	-	-
Reutebuch et al. (2015)	0.49	0.20	*	[0.09, 0.89]	1.22	No	1.94	0
Schatz (2017) (cloze)	0.61	0.27	***	[0.07,1.00]	-	-	-	-
Schatz (2017) (composite)	0.65	0.16	***	[0.35, 0.96]	1.22	No	2.47	1
Singh et al. (2017)	0.96	0.24	***	[0.48, 1.00]	4.78	Yes	2.47	3
Solis et al. (2021)	0.51	0.18	**	[0.16, 0.86]	0	No	1.89	1
Stringfield et al. (2011)	0.94	0.14	***	[0.62, 1.00]	4.56	Yes	2.61	4
Maintenance								
Chovanes (2018) (M)	1.00	0.48	***	[0.08, 1.00]	-	-	-	-
Head et al. (2018) (M)	1.00	0.29	***	[0.43, 1.00]	-	-	-	-
Howorth (2015) (Retell_M)	0.70	0.15	***	[0.40, 1.00]	-	-	-	-
Howorth 2015 (Composite_M)	0.24	0.16		[−0.09, 0.55]	-	-	-	-
Howorth et al. (2016) (Retell_M)	1.00	0.23	***	[0.54, 1.00]	-	-	-	-
Howorth et al. (2016) (Composite_M)	0.94	0.24	***	[0.48, 1.00]	-	-	-	-
Kim et al. (2018) (M)	1.00	0.24	***	[0.54, 1.00]	-	-	-	-
Schatz (2017) (cloze_M)	1.00	0.58		[−0.13,1.00]	-	-	-	-
Schatz (2017) (composite_M)	−0.40	0.28		[−0.95, 0.15]	-	-	-	-
Stringfield et al. (2011)	1.00	0.20	***	[0.62, 1.00]	-	-	-	-
Summary	**Estimate**	**SE**	**Significance**	**95% CI**				
Estimated total	0.67	0.07	***	[0.51, 0.83]				
Immediate effects	0.62	0.08	***	[0.44, 0.80]				

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TABLE 4
Tau-U, Visual Analysis, Estimated Total, and Moderator Analysis (*continued*)

Summary	Estimate	SE	Significance	95% CI
Maintenance effects	0.81	0.10	***	[0.66, 1.00]
Moderator analysis				
Intercept	0.55	0.15	**	[0.02.0.76]
Dosage	<0.001	0.01		[−0.01, 0.01]
Retell vs. Composite	0.34	0.17		[−0.04, 0.73]
Cloze vs. Composite	−0.31	0.12	*	[−0.58, 0.05]
Surface vs. Situation	−0.61	0.01	***	[−0.84, −0.38]
Textbase vs. Situation	0.02	0.14		[−0.29, 0.34]

Note. Q (df = 29) = 16.5177, $p > .05$; $\tau^2 = 0.00$ (SE = 0.05); $I^2 = 0.00$. FR = functional relation; Q/R = quality/rigor score on the SCARF (Ledford et al., 2020); Out = outcome rating on the SCARF.
*p<.05, **p<.01, ***p<.001.

than situation model level of instruction in reading comprehension instruction outcomes (i.e., surface vs. situation b = −0.61, SE = 0.10, $p < .001$, 95% CI = [−0.84, −0.38]). Textbase and situation model did not differ on reading comprehension outcomes ($b = 0.02; p > .05$). In group studies, both surface level and the textbase level of processing had significantly lower reading comprehension outcomes than instruction involving situation model level (i.e., surface vs. situation b = −0.19, SE = 0.06, $p < .05$, 95% CI = [−0.32, −0.05]; textbase vs. situation b = −0.16, $SE = 0.07, p < .05$, 95% CI = [−0.33, 0.00]).

Methodological Quality and Risk of Bias

SCARF for SSRD

We analyzed 15 designs across 14 studies using the SCARF framework (Ledford et al., 2020). Results of the SCARF analysis are provided in Figure 2. Given a possible range of scores on the SCARF of 0–4, the average quality/rigor score across the studies was 2.09 (SD = 0.51; range = 1.03–2.92). Using the recommended 1.7 as a cut-off score to indicate adequate quality to interpret the outcomes (Ledford et al., 2020), 12 of the designs (80%) met this criterion, with 8 of those 12 having a score of 2.0 or higher. Therefore, the majority of the included single-case designs had adequate to high quality and were deemed acceptable for interpreting outcomes. In examining the scatterplot (i.e., visual display of quality/rigor and outcome), although nine of the studies (60%) demonstrated strong outcomes (i.e., upper left and right quadrants), only five of those studies (33%) reported strong outcomes in a high-quality study (i.e., upper right quadrant).

Generalization and maintenance results are depicted in the bottom two scatterplots in Figure 4. Only five studies (33%) reported generalization measures with higher generalization outcomes depicted for pre–post measurement and lower generalization outcomes depicted when measured experimentally within the research design. Of the seven designs (47%) that reported maintenance data, the majority of the studies measured maintenance from 1 week to 1 month after ending the intervention. Moderate to high maintenance outcomes were found for these studies. However, based on the body of research, we cannot conclude whether reading comprehension intervention effects would be maintained more than 1 month following termination of the intervention (Figure 3).

WWC for Group Designs

The 12 group design studies included 12 intervention and two maintenance phases that were assessed for study quality. The majority (9) of the designs met WWC standards without reservations due to the randomization and low sample attrition. There was one study that met the WWC standards with reservations due to a high attrition rate but established baseline equivalence. Four studies did not meet WWC standards, and all of them were quasi-experimental design studies that failed to establish baseline equivalence. A decision flowchart is provided in Figure 4.

Risk of Bias

Results of the risk of bias evaluation are provided in Figure 5. Across the 12 group designs, 7 studies were judged to have some concerns in regard to potential bias in the study design and implementation. Only one study was judged as high risk (Turner et al., 2017) due to the randomization process. Areas of concern included procedures for randomization, evidence of adherence to the intervention, and measurement of the outcome variables. More specifically, the studies

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TABLE 5
Hedges's g and Moderator Analysis

Study	Hedges g	Variance	Sig.	95% CI
Arciuli and Bailey (2019)	0.32	0.16		[−0.46, 1.10]
Bailey et al. (2017)	0.09	0.19		[−0.76, 0.94]
Campbell (2010)	0.63	0.35		[−0.53, 1.79]
Engel and Ehri (2020) (Retell)	1.51	0.24	***	[0.55, 2.47]
Engel and Ehri (2020) (Composite)	0.14	0.18		[−0.69, 0.97]
Kamps et al. (2016)	0.08	0.06		[−0.40, 0.56]
Lemelman (2016)	0.14	0.10		[−0.48, 0.76]
Maximo et al. (2017)	0.62	0.14		[−0.11, 1.35]
Murdaugh et al. (2017)	0.25	0.15		[−0.51, 1.01]
Roux, Dion and Barrette (2015)	0.31	0.09		[−0.28, 0.90]
Roux, Dion, Barrette, Dupéré and Fuchs (2015)	0.49	0.28		[−0.55, 1.53]
Turner et al., 2017	0.29	0.13		[−0.42, 1.00]
Uccheddu et al. (2019)	0.14	0.36		[−1.04, 1.32]
Kamps et al. (2016) (M)	0.12	0.06		[−0.37, 0.61]
Roux, Dion and Barrette (2015) (M)	0.26	0.10		[−0.36; 0.88]
Moderator analysis	**Estimate**	**SE**	**Sig.**	**95% CI**
Estimated total	0.26	0.07	***	[0.13, 0.39]
Immediate effects	0.29	0.08	***	[0.13, 0.46]
Maintenance effects	0.15	0.10	***	[0.07, 0.24]
Moderator analysis				
Intercept	0.32	0.06	***	[0.18.0.47]
Dosage	<0.001	0.01		[−0.01, 0.01]
Retelling vs. Norm-referenced	1.34	0.06	***	[1.20, 1.50]
CBM Composite vs. Norm-referenced	0.17	0.10		[−0.09. 0.40]
Surface vs. Situation	−0.19	0.06	*	[−0.32, −0.05]
Textbase vs. Situation	−0.16	0.07	*	[−0.33, 0.00]

Note. Q (df = 14) = 9.31, $p > .05$; $\tau^2 = 0.00$ ($SE = 0.04$); $I^2 = 0.00$; M = Maintenance.
*$p < .05$, **$p < .01$, ***$p < .001$.

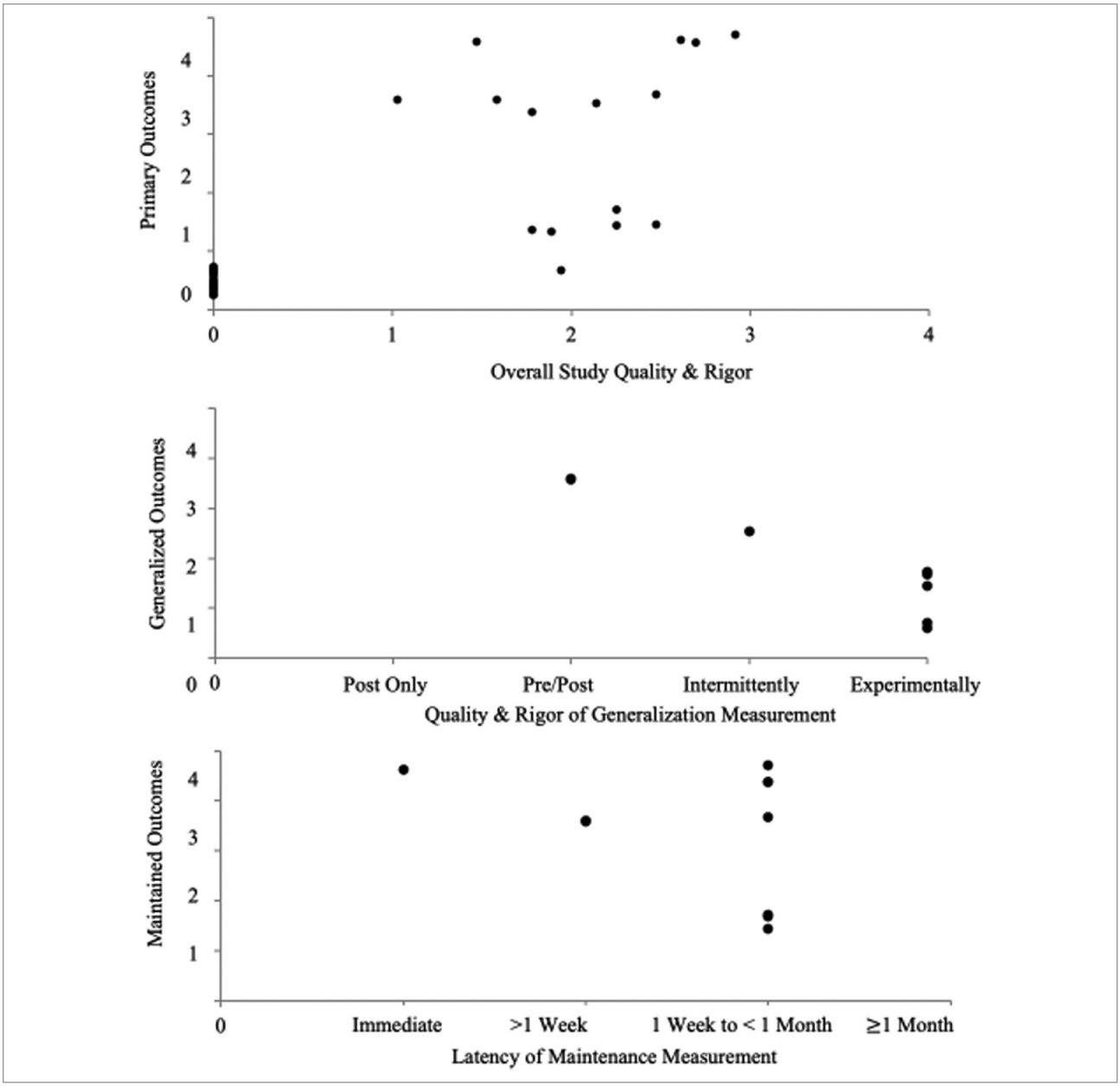
that were judged with some concerns did not provide evidence of procedural fidelity to ensure that interventions were implemented as intended and consistently did not blind the interventionist or the data collector to study condition assignment. Areas of strength across these studies include minimizing the effects of assignment on the intervention and reporting all outcome data. All SSRD studies were considered high risk for bias due to lack of blinding of participants, data collectors, and other personnel to study conditions. Areas of relative strength across these studies were inclusion of procedural fidelity measures and acceptable levels of reliability for measuring dependent variables.

Discussion

This systematic review and meta-analysis included 26 experimental studies (14 SSRD and 12 group design studies) of reading comprehension instruction for students with ASD. Anchored in Kintsch's (1988, 2019) C-I

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FIGURE 3
Results of the Quality Review for SSRD using SCARF (Ledford et al., 2020).



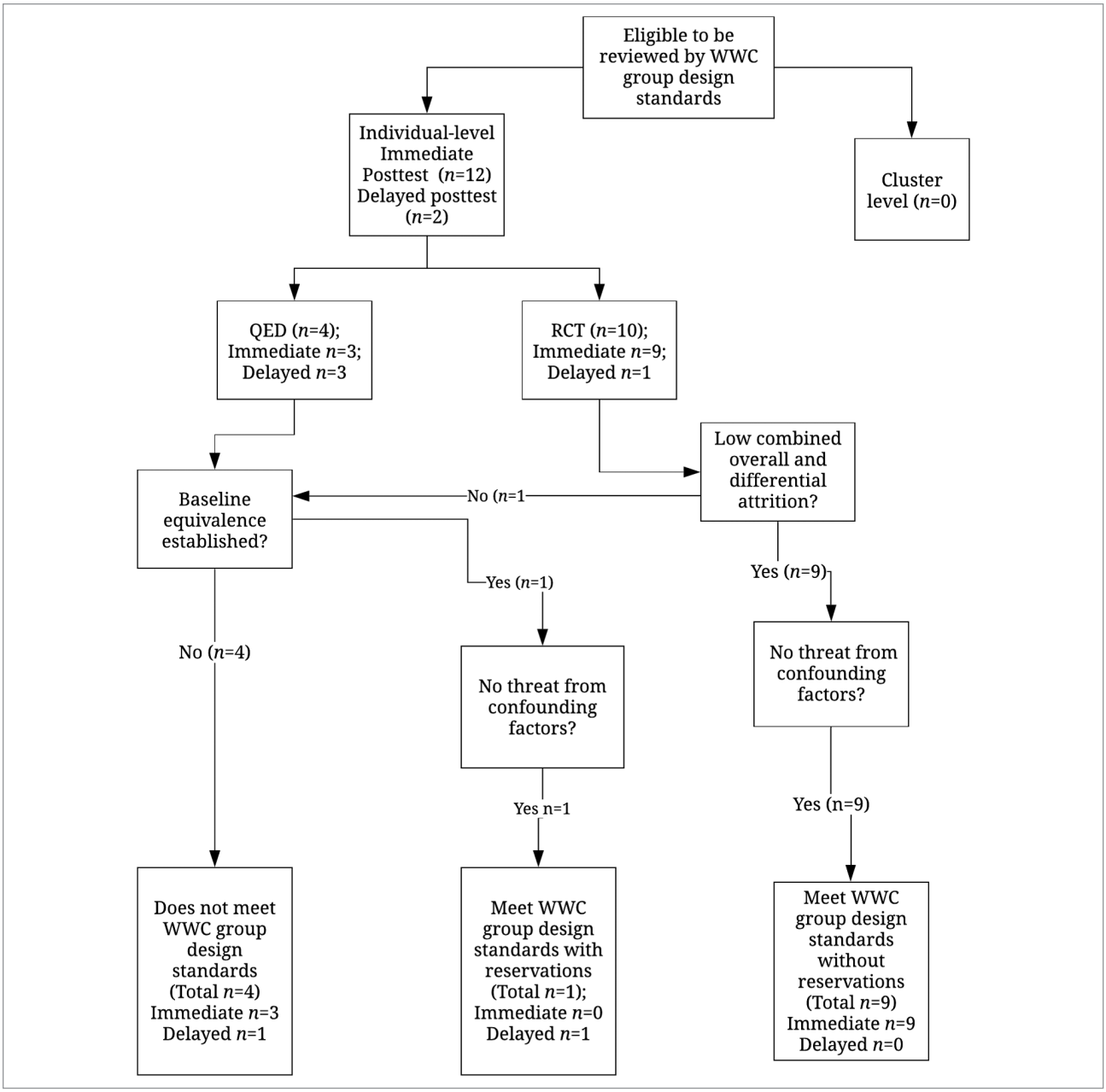
Note. Intermittently = during the course of a single-case design but with fewer than three data points in each condition. Experimentally = in the context of a single-case design with at least three data points per condition.

model of text comprehension, this review synthesized effect sizes for both SSRD and group design studies, providing a more complete examination of the intervention literature than previous reviews. SCARF and WWC standards for SSRD and group studies, respectively, were applied to allow clearer understanding of functional relations between the intervention and the reading comprehension outcomes. Risk of bias analysis was conducted to assess the quality of the interventions and their effects.

Our analyses generated several broad findings. First, most studies (n = 15; cf. surface n = 3; situation n = 8) addressed only the readers' processing of the textbase level of representation. Second, the SSRD studies had small immediate intervention effects (Tau-U = 0.62; g = 0.29), but the four SSRDs (Tau-U = 0.74) and four group designs (g = 0.33) addressing the readers' situation model construction demonstrated the most successful outcomes. Third, 80% of the SSRD studies were of adequate to high quality per SCARF guidelines, but only five studies demonstrated

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FIGURE 4
Results of the Quality Review Using What Works Clearinghouse (WWC, 2020)



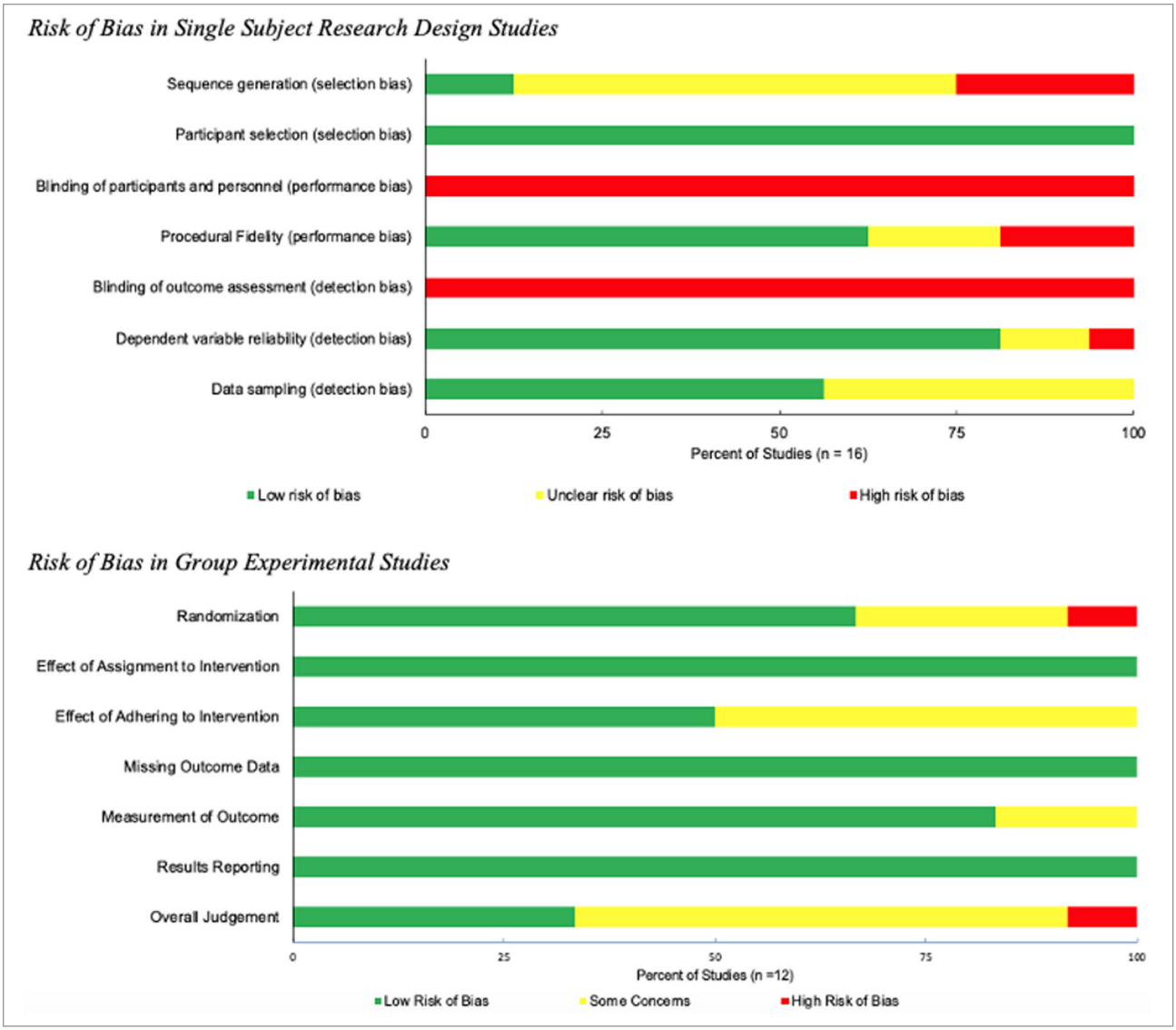
functional relations; similarly, 64% of group design studies met the WWC standards without reservations, but only one study had a statistically significant effect. Finally, the sequence generation, blinding, and outcome assessment procedures in SSRD studies showed high risks of bias.

RQ 1: What Level of Text Representation During the Intervention was Activated?

In the current review, eight studies (4 SSRD, 4 Group designs) actively engaged students with ASD in situation model building activities to enhance comprehension. For example, Schatz's (Schatz, 2017) instruction involved background knowledge activation and integration together with teacher-directed contextual application through script-based role-playing activities. Similarly, Lemelman (2016), Maximo et al. (2017), and Murdaugh et al. (2017) instructed students to picture the meaning of words, sentences, and paragraphs from narrative texts; the students were asked to give a summary based on their memories of the text and their imaging of meaning. Interventions such as these provide active attempts at integrating background knowledge with textbase understanding

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FIGURE 5
Risk of Bias For Ssrd (Upper) and Group Design Studies



to construct situation models for text. Alternatively, some studies, despite using inferential questions for their dependent variable assessments, did not provide scaffolds to actively engage students in background knowledge integration (e.g., Reutebuch et al., 2015). The processing level from these studies was considered textbase rather than situation model.

RQ 2: What was the Effect of Reading Comprehension Interventions?

We found that group design and SSRD studies that addressed situation model had better reading comprehension outcomes than studies focusing on textbase or surface-level instruction. Alternatively, just three studies applied surface structure reading (see Table 3) interventions aimed at improving reading comprehension. Our moderator analysis suggested, understandably, that these interventions demonstrated the lowest reading comprehension outcomes across SSRD and group designs. Although surface structure activities, such as matching interests of readers with text selection and repeated reading without comprehension scaffolding, provide essential foundations for reading comprehension (Kintsch, 1988), they require only minimal, surface-level engagement with text. For example, in this review, two surface structure interventions involved matching reading materials based on the restricted interests of students with ASD and having students practice reading fluency by reading aloud to therapy animals (Marshall, 2017; Uccheddu et al., 2019). Autism researchers might argue that WCC (Frith, 1989) accounts for minimal impacts on comprehension, but the

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C-I model suggests that the intervention focus on surface structure has little connection to comprehension. From either the C-I theory of reading comprehension or the theory of WCC in students with autism, it is evident that students need instruction and experience in constructing text meaning and integrating it into their world experience. We encourage future interventions to integrate surface model instruction and experiences with textbase level and situation model processing to deepen students' reading comprehension and their ability to both apply and generalize that comprehension to new and different texts and world experiences.

RQ 3: What was the Study Quality?

In the current study, we provided quality evaluations using SCARF and WWC for single-case and group design studies, respectively. Despite design factor differences, most studies had relatively high design quality, but few studies demonstrated functional relations. In SSRDs, eight (57%) studies received a quality score larger than two, but only five had outcome measures (Tau-U) larger than .93. In group studies, nine designs (64%) met the WWC group design standards without reservations, but only Engel and Ehri's (2020) retelling assessment showed a significant effect size.

Moreover, SSRD studies showed greater risk of bias than group designs. The former tended not to ensure a random sequence generation. For example, in multiple baseline designs, the sequence that participants received an intervention was seldom randomized, posing potential threats to internal validity. Two other factors posed threats to validity: blinding of participants and personnel (i.e., whether they knew they were in the baseline or treatment phases) and blinding of outcome assessments (i.e., whether data collectors knew what experimental phases the participants were in; Higgins & Altman, 2008). In fact, in the SSRDs, most measures were curriculum-based, non-norm-referenced measures such as retelling and short answer questions. As retelling and short answers demonstrated better reading comprehension scores than did norm-referenced assessments (see Table 5), the statistical validity of the results may be threatened (American Educational Research Association et al., 2014; Higgins & Altman, 2008). Future studies should further examine the reliability and validity of CBMs and incorporate multiple measures of reading comprehension.

Implications for Research

In the current review, reading comprehension was often assessed as composite measures (i.e., comprehension responses, sometimes integrated with background knowledge integration and sometimes not, were combined). Future studies should separate the outcomes on inferential questions from those of literal questions to allow examinations on whether the situation model level of instruction would exert more influence on inferential comprehension than literal comprehension. In addition, inferential questions that address textbase processing (e.g., main idea generation or inferencing word meanings based on context) should be separated from those that involve situation model level processing (e.g., questions that require background knowledge integration for prediction or conflict resolution). Furthermore, future studies should better identify which interventions are most appropriate for which subgroups of students with ASD by reporting measures of ToM, WCC, and EF, while aligning interventions more closely to the levels of text processing delineated in the C-I model. For example, students with lower levels of EF, who are known to have difficulty developing coherent understandings of text (Ozonoff et al., 2007) should respond well to interventions targeted at systematically linking microstructure propositions to form higher-order macrostructures. On the other hand, students with lower measures of ToM should benefit from interventions employing character maps to develop situation model representations of character motivation and actions.

Next, researchers should report reliability and validity information for curriculum-based reading comprehension measures and randomize participants' sequence of receipt of the intervention to improve study quality. As there was limited information across the studies on the overall IQ and reading performance and given the variability in these for individuals with autism, researchers are encouraged to report these data to ensure rigorous data analyses can be performed.

Moreover, few studies focused on building situation models and none directly compared instruction at two or more levels (e.g., textbase and situational models). Future research should explore which students with ASD benefit from what kinds of instruction at what levels of the C-I model. Only one group study demonstrated significant results despite relatively high overall study quality, whereas SSRD studies demonstrated medium effects, but only five studies (42%) suggested functional relations between the intervention and reading comprehension outcomes. Future studies should focus instruction in situation model building and improve design quality to investigate whether more functional relations between the situation model interventions and reading comprehension can be established.

Decoding skills ranged from 4 SDs below the average to 2 SDs above the average and the researchers tended to adapt the passage for reading intervention based on the decoding level of the students (e.g., Solis et al., 2021). Due to the influence of decoding in reading comprehension (Ricketts et al., 2013), the reading comprehension intervention outcomes might have been influenced by decoding skills. However, only 11 included studies reported decoding skills, which restricted the chance to study the impact of decoding on reading comprehension intervention. We

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encourage researchers to report complete decoding information and examine whether decoding is a moderator of reading comprehension intervention outcomes.

Implications for Teachers

Teachers need to be informed about the importance of addressing situation model building in their comprehension instruction. Comprehension tasks requiring background knowledge integration might be scaffolded before, during, and after reading. For example, after reading the first few pages of a narrative, students might engage in a summarization task that leads to a prediction about future text events. This process draws on their knowledge of general character action or story structure in combination with an understanding of the textbase.

Teachers also can help students visualize and contextualize meaning (Schatz, 2017). For example, teachers might support students in picturing meaning, creating dialogue between characters in a story, or situating themselves in the text by taking on different roles. Following such activities, students might retain relevant inferences in memory and develop coherent mental representations of text leading to a deeper understanding of textual meanings (Barth et al., 2015). In fact, these implications may serve a wide range of learner variability in any classroom setting, not just students with ASD (Barth et al., 2015; Pearson & Cervetti, 2015).

Limitations

The current study has several limitations. First, the small sample size in each study contributed to the difficulties of introducing more variables into moderator analysis. Second, the current review did not find significant heterogeneity among the effect sizes, so future studies should examine whether the reasons for these homogeneous effect sizes were indeed due to the small number of studies. Third, the current study only focused on reading comprehension, but we acknowledge that other components of reading, such as phonemic awareness, decoding, fluency, and vocabulary, also contribute to reading comprehension. Therefore, we encourage future meta-analytic studies to examine how these other components of reading may moderate reading comprehension intervention effects. Fourth, we relied on the authors' reports on the students' diagnosis of ASD, and future studies should validate these reports using criteria such as the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5).

Conclusion

The current research adopted a systematic and meta-analytic method to review reading comprehension instruction for students with ASD along with a methodology evaluation using multiple rigorous methods. Treatment effects across SSRDs and group design studies were small, and few studies demonstrated functional relations between the intervention and reading comprehension outcomes. Based on the meta-analytic results, further research is warranted that focuses on teaching students with ASD to create situation models for text. These studies and others should investigate how risk of bias affects interpretation of CBMs of reading comprehension. In addition, the research literature makes clear that students with ASD can and do learn to read with comprehension, that they are capable of processing text at multiple levels of text representation when provided supportive materials and instruction, and that both researchers and educators would better serve students' longer term learning needs by directing additional attention to instruction supportive of student creation of situation models for text. To provide more effective reading instruction to students with ASD, interventions should actively engage students not just in comprehending the textbase, but also in integrating background knowledge with that textbase into a larger understanding of the world.

Conflict of Interest

There is no conflict of interest in this research.

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SHUAI ZHANG, Assistant Professor, Department of Education, School of Arts, Sciences, Business, and Education, University of Saint Joseph (Connecticut), 1678 Asylum Ave, West Hartford, CT 06117, USA; email: shuaizhang@usj.edu

DEBRA A. PRYKANOWSKI, Assistant Professor, Reading Education and Special Education Department, Reich College of Education, 151 College Street, 325 E, Boone, NC 28608, USA; email: prykanowskida@appstate.edu

DAVID A. KOPPENHAVER, Professor, Reading Education and Special Education Department, Reich College of Education, 151 College Street, 338 B, Boone, NC 28608, USA; email: koppenhaverd@appstate.edu

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